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Khalifa University

Artificial Intelligence-Based Modelling and Optimization of the Green Methanol Production Process

Nabeel Sultan

MSc. Thesis

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A thesis submitted to Khalifa University of Science and Technology in accordance with the requirements of the degree of MSc. Chemical Engineering in the Department of Chemical & Petroleum Engineering.



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by

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at

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Abstract

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Technological advancements in machine learning (ML), Artificial Intelligence (AI), and data science have brought industries to the era of the fourth industrial revolution. The application of ML in chemical engineering is in the domains of process modeling, optimization, and predictive analysis. Traditional process modeling relies heavily on first-principal methods, which, while accurate, are computationally demanding and are non-flexible for variable process conditions. Renewable methanol produced through the power-to-liquid (PtL) process has gained significant popularity due to its various applications in household items, as a raw material for manufacturing valuable chemicals, and as a fuel both in blend or pure form. In today's competitive and uncertain chemical industry market, fast and accurate models are required to predict the plant output. This work aims to develop a data-driven surrogate model for methanol production process using ML to predict energy requirements, final product purity, and methanol production process. The effect of the sampling size and sampling technique (mainly Latin-Hypercube Sampling (LHS), Monte Carlo, and SOBOL) on the performance of the surrogate model is evaluated. A comparative analysis of different ML (e.g., XG-Boost, Random Forest, Decision Tree, Support Vector Regression) and Deep learning models (e.g., Artificial Neural Networks) is conducted using metrics such as coefficient of determination (R^2), mean-squared error (MSE), and mean-absolute-error (MAE). Additionally, the applications of these trained ML models are explored in optimization of process conditions to maximize production rate, enhance product purity, and reduce energy consumption rate. The results highlight that LHS performed significantly better than SOBOL and Monte Carlo techniques, and the surrogate model developed with the ANN technique shows superior performance with R^2 values of 0.9846, 0.9743, and 0.9750 for the energy, production rate, and final product purity predictions, respectively. Optimization results using the Differential Evolution Algorithm (DEA) reveal that the production rate can be increased by 33% with competitive purity but with an expense of 18% higher energy requirement from the base conditions.

Indexing Terms- Machine Learning, Artificial Neural Network, Surrogate Modeling, Methanol Production, Latin Hyper Cube

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Declaration

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List of Abbreviations

%AARD	Average Absolute Relative Deviation
AI	Artificial Intelligence
ANN	Artificial Neural Network
DEA	Differential Evolution Algorithm
DT	Decision Trees
FT	Fischer-Tropsch
GHSV	Gas Hourly Space Velocity
IEA	International Energy Agency
LHS	Latin Hypercube Sampling
MAE	Mean Absolute Error
MCS	Monte Carlo Sampling
MSE	Mean Squared Error
SMR	Steam Methane Reforming
NARX	Non-linear Autoregressive Network with Exogenous Inputs
NLP	Non-linear Programing
PCA	Principal Component Analysis
PTA	Power to Ammonia
PTM	Power to Methanol

R^2	Coefficient of Determination
RBF-NN	Radial Basis Function Neural Network
RF	Random Forest
RFF	Recycle Flow Fraction
RSM	Response Surface Methodology
SLSQP	Sequential Least Squares Programming
SMR	Steam Methane Reforming
SVR	Support Vector Regression
CCU	Carbon Capture and Utilization
DL	Deep Learning
GPR	Gaussian Process Regression
MAPE	Mean Absolute Percentage Error
ML	Machine Learning
NRTL	Non-random Two-liquid Model
PEM	Proton Exchange Membrane
PTG	Power to Gas
PTL	Power to Liquid
RBFN	Radial Basis Function Network
SVM	Support Vector Machine
VBF	Vanden Bussche-Froment Kinetic Model

Chapter 1. Introduction

1.1 Motivation

Climate change has been a primary concern in recent years due to increased carbon dioxide emissions and is the leading cause of global warming, which increases the earth's temperature and causes shifts in weather patterns all across the globe [1]. Rising climate concerns have forced developed countries to implement the Paris Climate Change Agreement to limit the global temperature increase to way below 2°C [2]. UAE has launched the Net zero emission strategy to align with the Paris Agreement [3], which calls on preparing a long-term plan to limit the rise in global temperature to 1.5°C by the end of 2050. For that, there must be a 50% drop in the total CO₂ emissions [4]. The current strategy to meet Net zero emission targets is to diversify the energy sources, decrease dependency on fossil fuels, and consider environmentally friendly options (i.e., solar and wind energy). The primary focus is on solar energy, which has experienced a reduction in the price of almost 70% in the last five years [5]. Solar is considered an intermittent energy resource, and increasing its content will significantly challenge maintaining grid stability. Therefore, grid systems require long and short-term energy storage options [6]. Two potentially potent ways for energy storage are power to methanol (PTM) and power to ammonia (PTA) [7], which utilizes the excess energy from the grid system to store it in the form of different energy carriers.

Methanol has gained significant popularity due to its various applications in household items (paints, varnishes, disinfectants) [8][9]. Its demand is also growing as a raw material for producing other industrial chemical products. Moreover, it can also be used as a fuel in the marine transportation sector in pure or as a blend with fossil fuels like gasoline [10]. Recently, the production of methanol from carbon dioxide captured from industrial sources and green

hydrogen produced using electrolysis has gained popularity as it provides an economical energy storage option for hydrogen and offers more sustainable solutions [11].

In today's competitive and uncertain chemical industry market, fast and accurate models are required to predict the plant output, which can help in real-time decision-making. The relentless advancements in technologies like machine learning (ML), artificial intelligence (AI), data science, and the Internet of Things (IoT) have brought industries into the era of the Fourth Industrial Revolution [12]. ML is essential in chemical engineering, mainly in process modeling, optimization, and predictive analysis. Traditional process modeling significantly depends on the first-principal methods, which demand the rigorous solution of differential equations for each piece of equipment. Though they provide accurate results, they are computationally intense and extremely non-flexible to accommodate variation in certain process conditions [13]. With the emergence of ML, AI has introduced robust, flexible models capable of describing and predicting the real-time behavior of chemical processes with a high degree of accuracy. These ML models are highly flexible and trained on large amounts of process data; therefore, they do not need an in-depth understanding of the process and complicated mathematics, thus making the whole process easier and more responsive [13].

In Chemical Engineering, process simulators like ASPEN and gPROMS play a significant role in accurately modeling industrial processes. These process simulators rely on derivative techniques; hence, the models developed using these simulators are highly non-flexible, sensitive to initial guesses, and the overall computational cost of solving these models is relatively high [14]. Additionally, using these models to optimize complex integrated processes is challenging due to their inherent derivative nature [13]. Therefore, a data-driven surrogate model could be a better alternative to describe the process behavior with high accuracy and speed [15]. These models are trained on large amounts of data from industrial processes and simulators to dynamically, adaptively, and expeditiously model and optimize the

process[14], [15]. Given the limitations of conventional process modeling methods, it is decided to develop an ML-based surrogate model for the renewable methanol production process and use these models in the multi-objective optimization to maximize the product purity and production rate while minimizing the process's total energy requirements.

1.2 Existing studies on methanol modeling using CO_2 hydrogenation

The area of process modeling of the methanol production process using CO_2 hydrogenation is actively explored by various researchers. A comprehensive literature review of the previous research is as follows:

Fereshteh et al. [16] studied carbon-capture integration and utilization for the methanol manufacturing process. This research work focuses on enhancing the conversion of CO_2 using an innovative three-stage reactor design approach to produce methanol over a commercial catalyst (CuO/ ZnO/ Al_2O_3). The catalyst is divided into three equal parts to load into the reactors. The exit stream from the reactors is passed through the flash drum to remove the liquid product from the gases, and the unreacted gaseous stream is passed to the next reactor. The findings reveal that the CO_2 conversion can be enhanced almost twice as much as the single stage. Although this research is theoretical and focuses on the technical feasibility of the process, it does not cover the economic aspects of this innovative process.

Borisut et al. [17] studied the process optimization of the methanol production process using the Aspen Hysys equilibrium reactor model. The research work aims to reduce the methanol production cost by applying response surface methodology (RSM) to optimize the reactor operating conditions that significantly impact the production cost. The data for the optimization using RSM is generated using 2^k factorial and Box-Benhken three-level design. The developed model shows good predictability with $R^2 = 0.87$ and an adjusted $R^2 = 0.86$. Optimization results

suggested a minimum production cost of \$565.54 per ton of methanol. Optimal conditions included operating pressures of 57.8 bar and 102.6 bar for the first and second reactor, temperatures of 183.6°C for the first reactor and 51.8°C for the first distillation column, and an outlet temperature of 63.5°C for the cooler after the second reactor.

Kiss et al. [18] presented a novel process for methanol production using a wet hydrogen stream from salt water electrolysis process. A continuous process is modeled using the Aspen Plus equilibrium reactor model, which utilizes the RK-Soave and NRTL property models for the reactor and separation sections. The proposed configuration uses a stripping column for the contact of wet hydrogen with a liquid mixture of methanol-water in a counter-current arrangement. This arrangement offers two advantages: it removes non-condensable gases from the methanol-water mixture and allows for complete recycling and it also simultaneously removes the water from the wet hydrogen stream, hence avoiding impact on reaction equilibrium conversation. It is also concluded that the proposed arrangement offers 15-20% energy savings and 5-8% more methanol production than conventional processes.

In another research, Borisut et al. [19] studied the optimization of reactor operating conditions for three different configurations to produce methanol using CO_2 hydrogenation, aiming to minimize the production cost. Three studied configurations include (1) a once-through reactor without recycling (2) a single-pass reactor with recycling, and (3) two reactors in series with recycling. Aspen HYSYS software's equilibrium reactor model is utilized to carry out this study, and an artificial neural network (ANN) model is used to map the reactor input-output relation. The results reveal that methanol production with two reactors in series has the lowest production cost. The optimum conditions are 70 bar and 199.04 °C for the first and 80°C for the second reactor. It is important to mention that this research work is based on an equilibrium

reactor model and did not factor in the operating variables of the separation process in the optimization of the final production cost.

Leonzio et al. [20] studied the optimization of reactor conditions for methanol production via CO_2 hydrogenation using Response Surface Methodology. The methanol production process is modeled using ChemCad 6.3 simulation software using the Graaf kinetic model. A central composite design is used to develop a mathematical optimization model for methanol production. Operating variables such as temperature, pressure, H_2/CO_2 ratio and recycling ratio are analyzed for their effects on methanol yield and reactor volume. The research highlights the significance of these factors in enhancing production rate while minimizing costs and volume. This research provides a detailed examination of operational conditions to optimize the performance of the methanol synthesis reactor.

In a research, Rafiee et al. [21] studied the hydrogenation of methanol with three objectives i.e., maximize the reactor productivity, CO_2 utilization and the conversion of hydrogen. This work includes the comparison of conventional methanol production on key performance indicators like CO_2 emissions. The results reveal that H_2 to CO_2 ratio of 3 results in 59% increase in the methanol production rate as compared to $H_2/CO_2 = 2$. This study also highlighted that the increasing H_2/CO_2 ratio increases natural gas consumption and eventually CO_2 emission in conventional processes, and if the hydrogen produced from the water electrolysis is employed, there will be negative CO_2 emissions.

Esmail et al. [22] explore the optimization of the methanol production process by examining various parameters such as catalyst type, temperature, pressure, and gas hourly space velocity (GHSV). By selecting an optimal catalyst and process conditions, the study investigates the impact of process modifications like hydrogen makeup injection, employing dual reactors, using inert gases, dry hydrogen, and recycling streams on methanol yield. The selected process

and catalyst were then simulated across eight scenarios, varying in parameters and recycle flow, and compared against a baseline to assess the optimization's effectiveness. The findings reveal significant improvements in operational costs because of the downsizing of equipment and reductions in recycle flow. Specifically, using a $\text{CuZnOAl}_2\text{O}_3$ catalyst led to a lower reactor temperature, a 38% decrease in recycle flow, and reduced electrical and steam consumption by approximately 5% and 67%, respectively. The optimized process enhances efficiency, reduces costs, and significantly lowers greenhouse gas emissions by 7526.35-ton of $\text{CO}_{2\text{eq}}$ from the methanol production process.

In another study conducted recently by Wolday et al. [23] a multiobjective optimization of methanol using syngas as a feed is demonstrated using Aspen plus model integrated with Matlab NSGA-II algorithm. The main aim of this study was to enhance the total process utility requirements and to reduce the annual GHG emissions by adjusting the process variables of the separation section. The integration of matlab optimization algorithm directly with the Aspen Plus simulation required the software to calculate the physical properties of species and solve various differential equations in each iteration. Therefore, the computation cost of solving these models was considerable, ranging from 3945 to 5239 minutes for different cases, as reported in this study. The results of optimization demonstrated significant improvements, including an 8.595% increase in methanol production rate and a remarkable reduction of 99.973% in total waste generation.

1.3 Significance of current work

Most of the previous research work focuses on the methanol surrogate model development from CO_2 hydrogenation using RSM and factorial designs [17], [20] with an equilibrium reactor [17], [19] or Graaf kinetic model [22]. RSM has been utilized for developing surrogate

models for a very long time with high efficiency for small sets of variables with acceptable accuracy. However, it loses its performance as the number of variables increases and if the functions are nonlinear [24]. On the other hand, the previous research works are based on the equilibrium reactor model [17], [19], which can not represent the real-world output of a process. It has also been concluded in various research studies that the Graaf kinetic model is unable to capture the actual process behavior under variable temperature-pressure conditions of reactors and gives abnormal trends [25], [26]. Therefore, current work focuses on utilizing Vanden Bussche-Froment (VBF) kinetics in preparing ML-based data-driven surrogate models for the methanol production process from direct CO₂ hydrogenation. The developed models can be integrated with multi-objective optimization and can improve the overall computational time requirements of the optimization. Furthermore, this research also investigates the influence of different space-filling sampling techniques used for generating datasets (Latin-Hypercube Sampling (LHS), SOBOL, and Monte Carlo) and sample sizes on the performance of the surrogate models. This research also tends to utilize the developed model in multi-objective optimization of the overall process to maximize the production and purity of the final product while minimizing the total energy requirements. Developing an ML-based data-driven surrogate model for process will also offer following advantages:

- 1. High Accuracy and Interpretability:** ML-based surrogate models can give fast prediction of plant output with high accuracy. The models can learn complex non-linear relationships within the data and can be more interpretable, providing insights into the factors affecting the process. Their ability to provide accurate and fast predictions can also help plant operators and decision-makers to make informed and planned decisions in the Pareto frontier.
- 2. Low Computational Cost:** Compared to first-principal models, the computational cost of an ML-based surrogate model is relatively low and can assist in real-time decision-

making to maximize the output of the process. Low computational cost also enables efficient exploration of different process configurations, parameters, and operating conditions that contribute to the overall efficiency of the optimization process.

3. **Integrated process optimization:** Methanol production plants' integration with carbon-capturing and renewable energy resources can bring new integrated process optimization challenges, i.e., the operation decision of one plant may impact the other. ML-based surrogate models can facilitate fast and real-time optimization of the integrated process with high accuracy due to its flexibility to incorporate uncertain conditions.

1.4 Thesis Objectives

The main objectives of the thesis are summarized as follows:

- Simulate the methanol production process on ASPEN plus with VBF kinetic model by utilizing the captured carbon dioxide and hydrogen produced from electrolysis.
- Generate data sets from the Aspen Plus simulator with different sample sizes using space-filling sampling techniques (Latin-Hyper-Cube (LHS), SOBOL, and Monte-Carlo).
- Analyze the effect of sampling techniques (LHS, SOBOL, and Monte-Carlo) and sample sizes on the training of machine learning models.
- Develop ML-based surrogate models for the methanol production process capable of fast and accurate prediction of energy requirement, product purity, and production rate under variable operating conditions.

- Utilize the developed surrogate model to determine the optimum operating conditions for maximizing the methanol production rate and purity while minimizing the total energy requirements.

1.5 Thesis Outline

This thesis is organized in the following format:

Chapter-2 covers an extensive literature review of the topic. An overview of power-to-liquid technologies, surrogate modeling methods, sampling techniques, and relevant concepts of ML and deep learning models are provided in this section.

Chapter-3 covers the research methodology and framework employed in this research. It includes a simulation of the methanol production process, techniques utilized for data generation, as well as ML and deep learning models chosen for the surrogate modeling.

Chapter-4 comprises the findings derived from the results and provides an in-depth analysis. A comparison of different sample sizes and sampling techniques on the performance of the selected model is presented, along with a comparison of different ML and DL-ANN model performances for predicting plant production rate, purity, and energy requirements. This section also includes the optimization results to improve the overall efficiency of the process.

Chapter-5 includes the overall conclusion drawn from the research work along with possible recommendations based on insights gained and future work direction.

Chapter 2. Literature Review

2.1 Industrial Sources of CO₂

2.1.1 CO₂ from the Cement industry:

Cement consumption in a country is considered an essential factor in its economic growth. It plays a vital role in the construction sector and serves as a building block for residential and commercial infrastructure development. According to the International Energy Agency (IEA), the cement industry is the third most significant contributor of CO₂, accounting for 8% of the total GHG emissions [27]. Several energy-intensive processes are involved in its manufacturing, including extraction of raw materials, processing, and clinker formation [28].

CO₂ emissions from the cement industry can be categorized into two stages. The first stage is the chemical reaction involved in clinker formation, which is the primary component of the cement. It comprises decomposition of limestone (CaCO₃), by adding heat to produce calcium oxide (CaO) and CO₂. The second stage includes the CO₂ emission from the fossil fuel combustion used to heat the reactant at well above 1000°C and the emission associated with the purchased electricity. **Fig.1** shows the sources of CO₂ emissions from a cement plant.

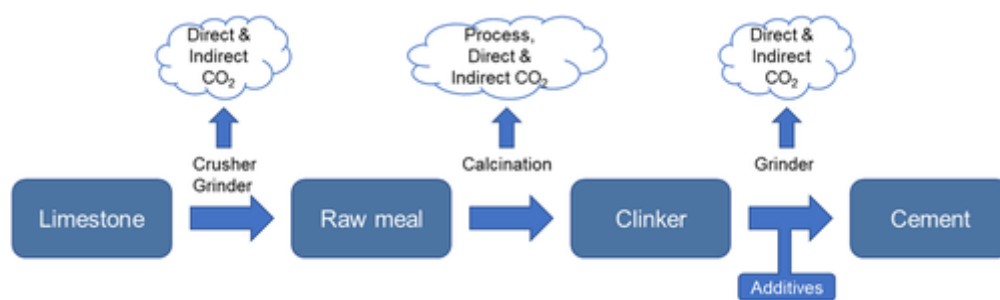


Figure 1: Cement production process and CO₂ sources [29]

The composition of flue gases from the cement plant depends on the nature of the fuel used in its manufacturing. A variety of fossil fuels are used in its manufacturing, including natural gas, oil, petroleum coke, coal, and different solid waste materials [28]. Due to significant CO₂

emissions, the capturing plant installation is essential and serves as a high-purity CO₂ stream source, which can be used for sustainable fuel production.

2.1.2 CO₂ from the Steel industry:

Most steel manufacturing plants use an integrated process in which iron ore is first converted into crude iron in a blast furnace (BF) by utilizing coke as a reducing agent; and after that, the crude iron is melted in a basic oxygen furnace (BOF) to produce steel [30]. An additional process melts the pig iron and scrap metals to produce steel in an electric arc furnace [31]. Steel manufacturing is an extremely energy-intensive process and significantly contributes to CO₂ emissions. The whole manufacturing process of steel and possible CO₂ sources are explained in *Fig.2*.

Fig.2.

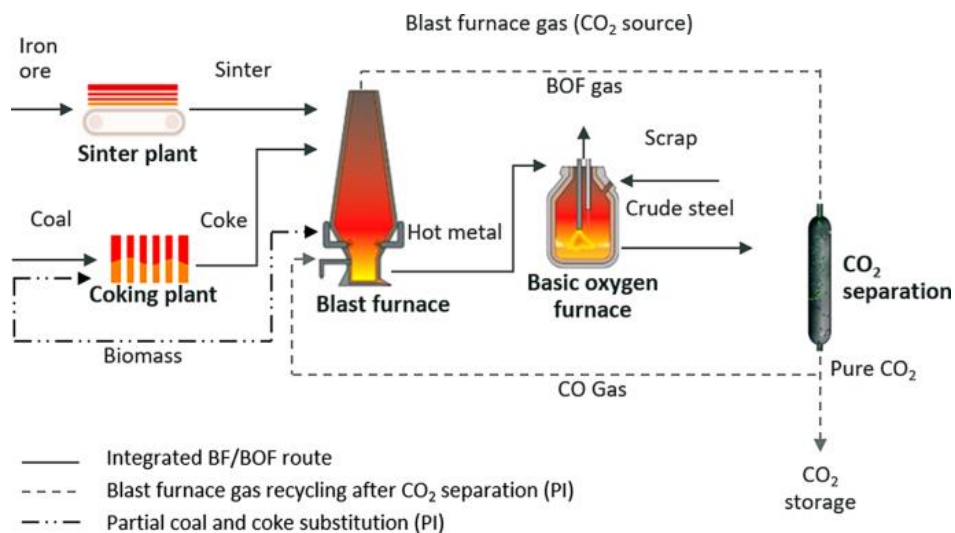


Figure 2: Steel production process and CO₂ sources [32]

Various methods have been introduced in the last few years to reduce GHG emissions from steel production, which are classified into two groups. The first approach involves smart carbon usage, in which CO₂ emissions from steel mills are captured and either utilized in the oil recovery or stored inside underground geological formations [33]. The second approach is called Carbon Direct Avoidance, in which the process of steel manufacturing is modified inherently to

avoid emissions. This approach includes recycling carbon monoxide in the blast furnace or the utilization of H_2 instead of Coke as a reducing agent [34].

A typical lifetime of the steel industry is about 30-50 years [35]. Suppose the technology of iron ore reduction by using green hydrogen becomes feasible. In that case, it will not be possible to completely shift the whole industry to a new process within the given time frame. The forecast indicates that by the end of 2050, about 50% of the steel industry will feature the Carbon Avoidance process [36]. Limiting climate change to acceptable levels and meeting the sustainability target carbon-capturing utilization and storage will play a significant role in reducing GHG emissions by 2050.

In current practices, the captured flue gases are stored inside underground geological formations or utilized for oil recovery, and they provide little economic value. It could be a valuable option to convert the CO_2 captured stream to produce renewable methanol, which can be processed further to produce a variety of hydrocarbon fuels.

2.2 Power-to-X (PtX) Technologies:

PtX is an innovative concept that provides pathways to utilize the excess renewable energy from the grid systems and carbon dioxide captured from various sources to convert it into versatile energy storage carriers. The main component of this technology is to produce Hydrogen and to store it in valuable chemicals. **Fig.3** represents a general process description of utilizing excess energy and captured carbon dioxide to produce various valuable products.

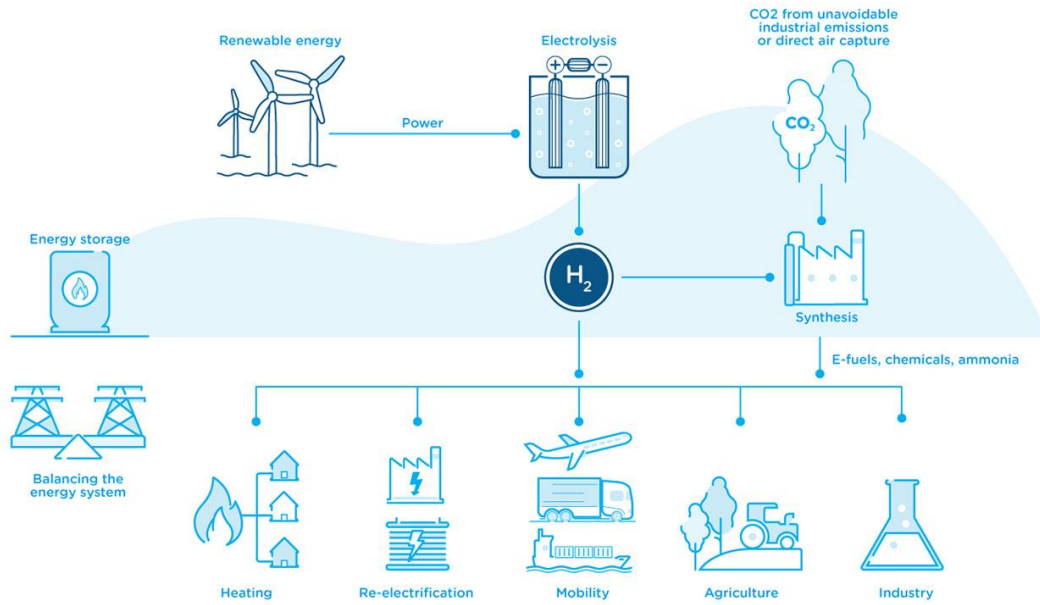


Figure 3: Power-to-X concepts for different energy storage carriers [37]

2.2.1 Hydrogen Production Methods

Hydrogen can be produced from various sources and methods, including renewable and non-renewable pathways. Most of the hydrogen (95%) as of 2020 is made from non-renewable resources by utilizing fossil fuel in the steam reforming process and emitting about 830 million tons/year of CO₂. The rest of the hydrogen is produced from the electrolysis process using renewable energy [38]. The primary hydrogen production and consumption pathways are shown in **Fig.4**.

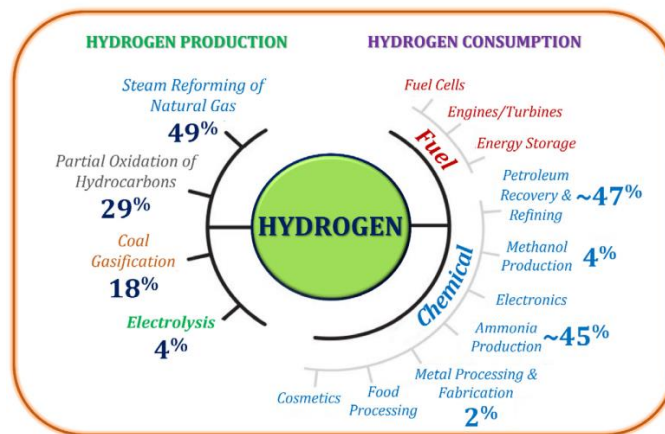


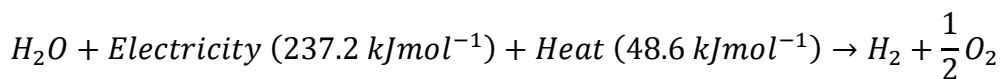
Figure 4: Major Hydrogen production and consumption pathways [39].

Hydrogen can be classified into different color schemes depending on the production technology and energy resources, i.e., gray, blue, brown, green, and black. **Fig.5** gives a complete insight into comparing various hydron production technology and key parameters.

Hydrogen Color	Technology	Source	Products	Cost (\$ kg/H ₂)	CO ₂ emissions
Brown Hydrogen	Gasification	Brown coal (Lignite)	H ₂ + CO ₂	1.2–2.1	High
Black Hydrogen	Gasification	Black coal (Bituminous)	H ₂ + CO ₂	1.2–2.1	High
Grey Hydrogen	Reforming	Natural gas	H ₂ + CO ₂ (Released)	1–2.1	Medium
Blue Hydrogen	Reforming + carbon capture	Natural gas	H ₂ + CO ₂ (Captured 85-95%)	1.5–2.9	Low
Green Hydrogen	Electrolysis	Water	H ₂ + O ₂	3.6–5.8	Minimal

Figure 5: Classification of hydrogen based on production source along with key features [39].

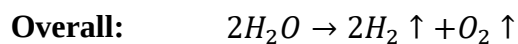
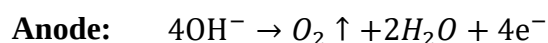
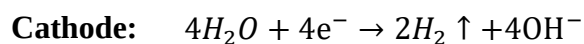
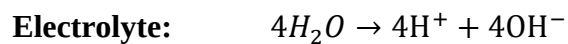
Green hydrogen has proven globally to be a promising fuel for sustainable development. It can be produced from spitting water molecules by utilizing renewable energy like solar and wind, which are emission-free technology [40]. It is an endothermic process, requiring energy in the form of electric current through an electrochemical cell. The electrochemical cell can be connected in parallel or in series arrangement to increase the overall production and efficiency of the process. The main reaction of water electrolysis is as follows:



This reaction requires a constant source of 1.23V theoretical dynamic cell voltage to split the water molecule into oxygen and water [39]. There are four types of water electrolysis technologies that are currently available as commercial products depending on their electrolyte, ionic agent (OH⁻, H⁺, O₂⁻), and their operating conditions, such as (i) Alkaline water electrolysis (ii) AEM water electrolysis (iii) PEM water electrolysis and (iv) Solid water electrolysis.

2.2.1.1 Alkaline water electrolysis:

Alkaline water electrolysis is a well-matured and reliable technology for producing hydrogen on an industrial scale for various commercial applications all around the world [39], [41]. It was first introduced by Troostwijk and Diemann in 1789, and the first alkaline water electrolyzer plant went into operation in 1939 with a production capacity of $10,000 \text{ Nm}^3\text{H}_2 \text{ h}^{-1}$ [42]. In an alkaline electrolyzer, the electrodes are made up of Nickel (Ni) coated stainless steel material, and asbestos/ ZrO_2 -based diaphragm separators are used [39]. The electrolyte used in this electrolyzer is basic; mainly, KOH or NaOH is primarily used [43]. When suitable potential is applied between the electrodes, the water molecule splits into H^+ and OH^- ions. This splitting triggers two individual half-cell reactions, such as oxygen production at the anode, called the oxygen evolution reaction (OER), and hydrogen production at the cathode, called the hydrogen evolution reaction (HER). During the half-reactions, two moles of alkaline solution are reduced to produce one mole of hydrogen gas, which is removed from the cathodic surface, and two moles of hydroxide are produced, which is transferred in the internal circuit under the influence of an electric circuit and reach the anode, where the hydroxyl ions (OH^-) are discharged to produce a half mole of oxygen gas which is removed, and one molecule of water is produced. The whole schematic of alkaline water electrolysis is shown in **Fig.6**. The main challenges associated with the alkaline technology is the low current density due to the mobility of hydroxyl ion and the corrosive nature of the electrolyte used in the process (KOH) [43]. The reaction involved in the alkaline water electrolysis process is summarized as follows:



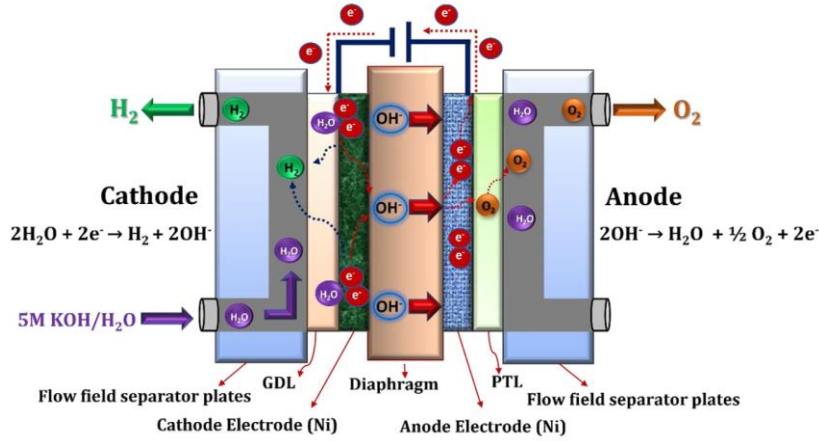


Figure 6: Alkaline water electrolysis working principle [39]

2.2.1.2 Polymer Electrolyte Membrane (PEM) Electrolysis:

PEM-based membranes are designed to overcome the limitation of the alkaline water electrolysis processes [39], [41]. The PEM electrolysis process was introduced by Grubbs and first developed by General Electric Co. in 1966 [41]. The PEM water electrolysis process is similar to the PEM fuel cell, where the sulfonated polymer membrane is used as an electrolyte [39]. In PEM, the ionic charge carrier is a solid proton instead of a hydroxide ion, like in an alkaline exchange membrane [39]. The PEM water electrolyzer operates at a low temperature and has a high current density [41]. High purity of gases can be obtained in PEM electrolysis. The PEM technology is safer than alkaline water electrolysis due to the absence of a caustic electrolyte [39]. Based on high efficiency and advantages, much of the water electrolyzer is manufactured based on PEM water electrolyzer for transportation and industrial applications.

In the PEM water electrolysis process, water molecule splits into hydrogen and oxygen electronically. When an electric voltage greater than the thermoneutral voltage is applied, the water molecules at the anode side decompose to produce oxygen, proton, and electrons. Oxygen is removed from the anode surface while electron and proton are transferred to the cathode side through the proton conducting membrane, where they combine to form the hydrogen gas. The main challenge in PEM electrolysis technology is the high cost of the

component, which include collector, electrode, and bipolar plate materials [39]. The whole schematic of PEM water electrolysis is shown in **Fig.7**, and the reaction involved in the process are summarized as follows:

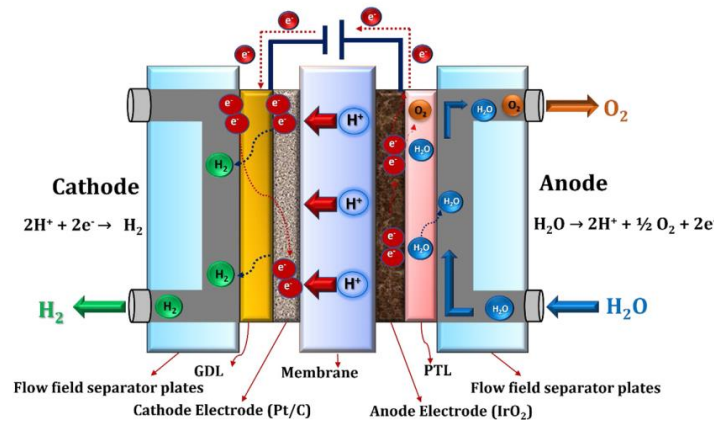
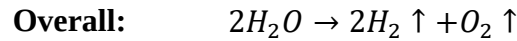
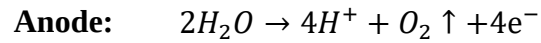
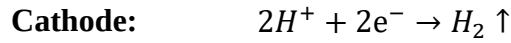


Figure 7: PEM water electrolysis working principle [39]

2.2.1.3 Anion exchange membrane (AEM) electrolysis:

Unlike alkaline electrolysis and PEM, AEM water electrolysis is a developing technology for green hydrogen production[39]. It is also known as alkaline-PEM; the combination of both alkaline and PEM electrolyzers and overcomes their weaknesses. It contains a thin anion membrane that serves as an electrolyte, reducing the operating cost of using the proton exchange membrane as in the PEM electrolyzers. It uses cheap transition metal catalysts instead of noble metals and low concentrated 1M KOH alkaline solution as compared to 5M KOH solution required in conventional alkaline electrolysis. The electrochemical reaction in AEM water electrolysis consists of two half-reactions similar to an alkaline water electrolyzer; at the cathode, the water molecule is split and reduced to generate H_2 and hydroxyl ions with the addition of two electrons [39]. Hydrogen is removed from the surface of the cathode, and hydroxyl ions are diffused through the ion exchange membrane due to the attractive force

created by the anode, while the electrons are transferred through an external circuit and reach the anode side, where hydroxyl ions and electrons are combined to form oxygen and two electrons. AEM technology promises various advantages but is in the developing stage and still requires improvement in stability and cell efficiency. The oxygen gas is removed from the surface of the anode. The whole schematic of AEM water electrolysis is shown in **Fig.8**, and the reaction involved in the process are summarized as follows:

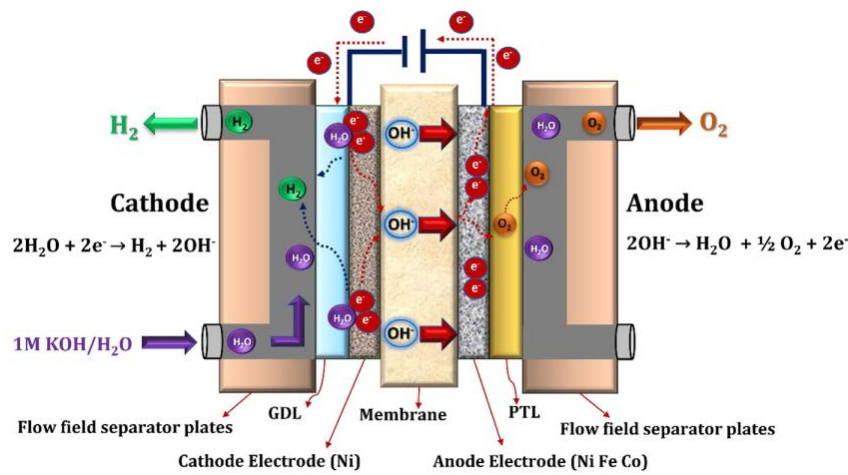
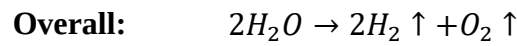
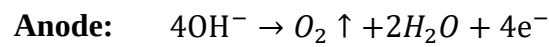
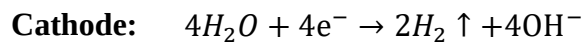


Figure 8: AEM water electrolysis working principle [39]

2.2.1.4 Solid oxide water electrolysis.

Solid water electrolysis is another method of producing green hydrogen. It is first developed in the USA in the year 1970s [44]. Solid oxide water electrolysis operates with water in the form of steam at a high temperature of about 500 – 800 degrees [39], [44]. It can drastically reduce the power consumption for water splitting by increasing the efficiency of the process[39]. In this process, the water molecules at the cathode side are reduced to hydrogen and oxygen ion O^{2-} with the addition of two electrons [39]. The hydrogen is released from the cathodic surface, and the oxide ion is traveled through the ion exchange membrane to the anode side. At the

anode, the oxide ion is converted into oxygen gas by reducing further with the removal of electrons. The oxygen gas is released from the anode surface, and the electrons are again traveled from the anode to the cathode side through an external circuit. The whole schematic and working principle of solid oxide water electrolysis is shown in **Fig.9** and the reaction involved in the process are summarized as follows:

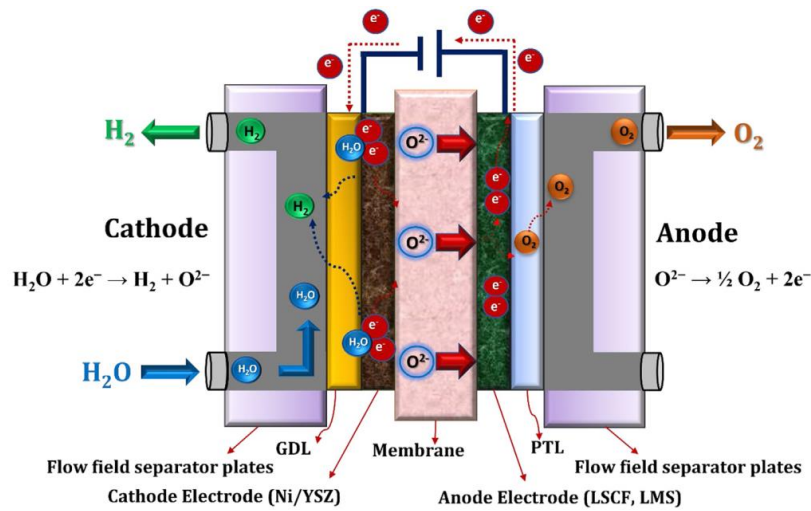
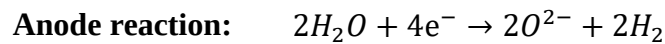


Figure 9: Solid oxide water electrolysis working principle [39]

2.2.2 Power-to-Liquid (PtL) and Power-to-Gas (PtG).

Among many diverse applications, PtX technology can be categorized into two main pathways i.e., Power-to-Liquid (PtL) and Power-to-Gas (PtG). The power-to-gas concept involves converting renewable energy resources such as solar and wind into gaseous fuels, primarily hydrogen and methane. The electrolysis process is used for the production of hydrogen, which can be directly utilized as a clean fuel or further converted into other chemicals by combining it with carbon dioxide, such as methane, offering a temporary and transportable form of renewable energy. Methane and hydrogen can also be easily injected into the natural gas infrastructure to lower emissions and meet sustainability targets [45]. Power-to-liquid technology works similar

to power-to-gas, but instead of producing green hydrogen or renewable natural gas, it produces liquid fuels like methanol, ammonia, diesel, and aviation fuels as a final product. It includes converting the CO₂ stream captured from industrial processes like cement, steel, power generation, and bio-gas plants to liquid hydrocarbon fuels [46]. The main difference between the PtL and PtG processes is the technology used in the CO₂ hydrogenation section; in the case of PTG, the CO₂ stream is converted into renewable natural gas by the process called Methanation, also known as the Sabatier reaction. In the case of the PTL process, direct hydrogenation or Fischer-Tropsch (FT) synthesis is employed to convert CO₂ into liquid hydrocarbon fuels. Various conversion pathways have already been explored, and some of them are explained in the **Fig.10**.

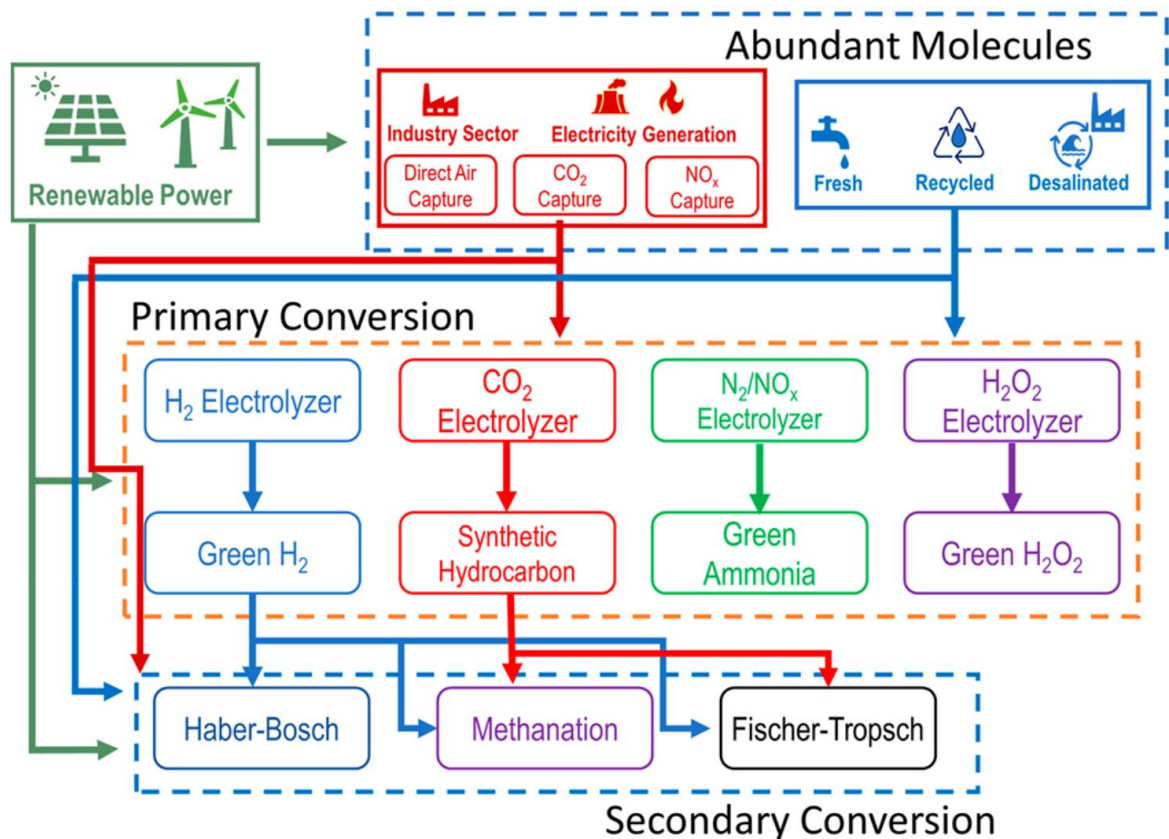


Figure 10: Schematics of various PtX conversion pathways [47]

Hydrogen demand produced through PtX technologies is increasing as it plays an important role in transitions towards sustainable energy systems. The current challenge in utilizing

hydrogen as a fuel is the issues associated with its storage. Hydrogen is a low energy-density molecule; it requires low temperature and high-pressure conditions to be stored. Hydrogen transportation will also be expensive due to special storage and handling requirements [48]. Therefore, utilizing hydrogen to produce liquid fuels could be a profitable and environmentally friendly option to meet sustainability targets. Two potentially potent candidates for hydrogen storage are power to methanol (PTM) and power to ammonia (PTA) [7], which utilizes the green hydrogen produced through the PtX concept to store it in the form of different energy carriers.

Methanol has gained significant popularity due to its various applications in household items (paints, varnishes, disinfectants) [9]. Its demand is also growing as a raw material for manufacturing other industrial chemical products. Moreover, it can also be used as a fuel in the marine transportation sector as a blend with fossil fuels like gasoline [8]. The conventional methanol production process relies on the steam methane reforming (SMR) process. It uses natural gas as a feedstock, posing environmental challenges due to associated carbon dioxide emissions. Producing methanol by utilizing green hydrogen and captured carbon dioxide will be a promising avenue to decarbonize the industrial sector and store hydrogen in methanol. This approach also helps in lowering the overall carbon footprint. **Fig.11** shows the block flow diagram of the methanol production process using conventional SMR and carbon-capturing and utilization approach.

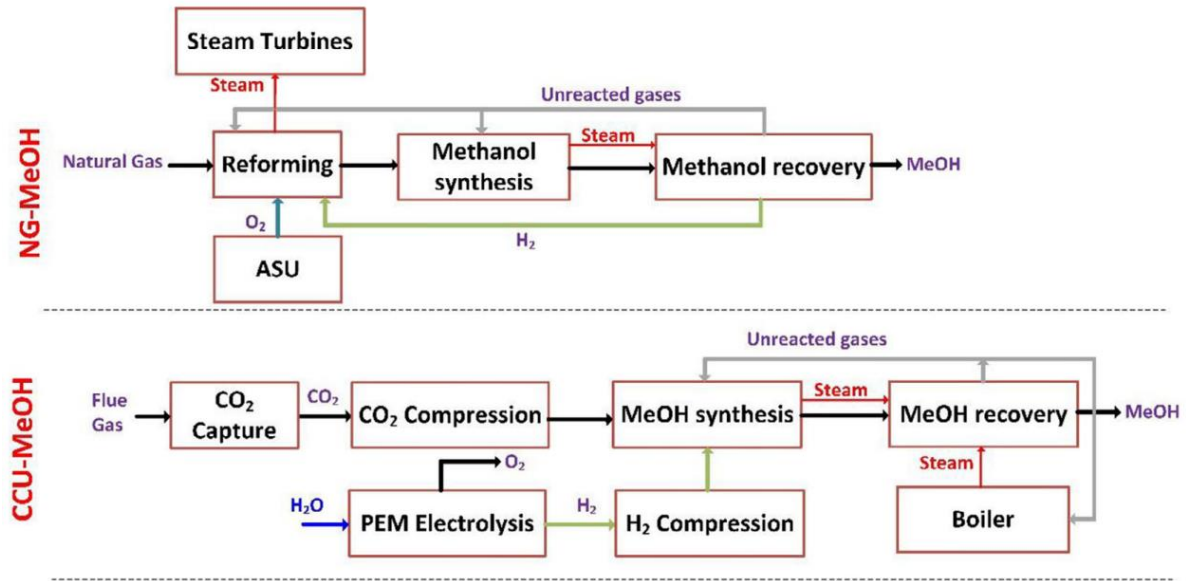


Figure 11: Block flow diagrams of the two methanol production pathways [49]

2.2.3 Comparison of different PtX pathways

A comprehensive comparison of different products that can be produced using PtX technologies is given in *Table.1*

Table 1: Comparison of different energy carriers from PtX technologies

Feature	Hydrogen	Liquid Ammonia	Renewable Methanol
Boiling Point	-252.9 °C at 1 atm [50]	-33.34°C at 1atm [50]	Liquid at room conditions [51]
Energy Density	8.5 MJ/L (Liq) [50]	12.7 MJ/L (Liq) [50]	15.8 MJ/L [50]
Storage Conditions	Cryogenic conditions (P>350 bars) [50]	Pressurized (P≈ 10 bars) or refrigerated at ambient pressure [50]	Stored in the conventional fuel tank at ambient conditions [51]
Transportation Conditions	High-pressure tanks and cryogenic liquid tankers [52]	Pressurized or refrigerated tanker for liquid transport [53]	Conventional fuel tankers, no special storage requirement [51]

Environmental Impact	Low operational emissions; challenges include water usage for electrolysis [52]	Produce no direct carbon emissions; challenges include toxicity and safe handling [53]	Potential for carbon neutrality when produced from CO ₂ and renewable energies [51]
Scalability	High, with advancements in electrolysis and storage technology [52]	Moderate to high; limited by demand and safe storage [53]	High, especially with biomass feedstock; requires further development in CO ₂ capture [51]
Safety Considerations	Highly flammable and low density requires careful handling; non-toxic. [52]	Toxic and hazardous, require special safety measures for storage; risk of ammonia exposure to humans and environment [53]	Lower toxicity and flammability as compared to others, making it safer in handling and transportation [51]
Lethal Dosage LC50 for fish population	N/A	0.068 mg/L [50]	15,400 mg/L [50]

2.3 Surrogate Modeling

Surrogate modeling is defined as developing a functional relation between input and output variables [12], [54]. Process modeling simulators like Aspen play an essential role in the accurate modeling of chemical processes. However, these models rely heavily on derivative techniques and are computationally intensive. Surrogate models are frequently used in chemical industries to minimize computational time and for optimization [14]. A systematic, iterative approach to developing a surrogate model is shown in **Fig.12**. It involves selecting a suitable design of experiment technique, data acquisition from the simulator, model identification, training, testing, and validation of the model [55].

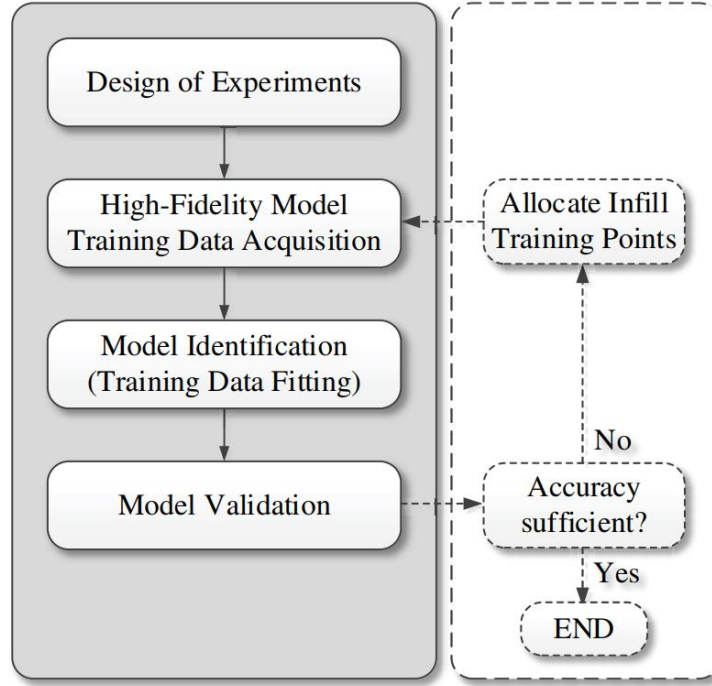


Figure 12: Surrogate model development process [55]

There are various methods used for developing accurate surrogate models, some of them are as follows:

2.3.1 Response surface methodology (RSM)

RSM has been utilized for developing surrogate models for a very long time, with high efficiency for small sets of variables an acceptable accuracy [24]. The RSM models can be built by fitting the data to linear or polynomial functions depending on empirical observations and considering the inherent characteristics of the data. The main goal of RSM is to find an optimal response (y), influenced by several independent variables (x)[56]. The mathematical formulation of 2nd-order RSM model is as follows:

$$y = \beta_0 + \sum_{i=1}^k \beta_i x_i + \sum_{i=1}^k \beta_{ii} x_i^2 + \sum_{i < j} \beta_{ij} x_i x_j + e$$

Where y is the response, x_i and x_j are the independent variables, β s are the adjustable coefficients, and e represents the error.

Factorial designs are most commonly used in RSM to plan the number of experiments that cover the region of interest in the variable space. It includes a Center Composite Design or Box-Behnken Design. RSM is a very convenient surrogate modeling method that gives fast, acceptable predictions but loses its performance as the number of variables increases and if the functions are nonlinear [24].

2.3.2 Taguchi Method

The Taguchi method is a robust method widely used to optimize industrial processes and product designs. It uses an orthogonal array design to structure the experiments that cover the multi-variable space with the minimum number of experiments [57]. Unlike RSM, the mathematical formulation of the Taguchi Method does not fit the data as a model. Instead, it uses a signal-to-noise (S/N) ratio, which measures the sensitivity of the quality characteristic being investigated to the factors that cause variation, e.g.

$$S/N = -10\log_{10} \left(\frac{1}{x} \sum_{i=1}^n \frac{1}{y_i^2} \right)$$

Where x is the number of observations, and y_i represents the observed values. The main aim here is to maximize the ratio, which eventually minimizes the variability around the target variables [57]. The use of orthogonal arrays significantly reduces the number of experiments, increasing the overall robustness of the process, but the model's accuracy is compromised. Therefore, this technique is suitable for initial screening and analysis of the process.

2.3.3 Support Vector Regressor (SVR)

The SVR model is a robust and efficient ML model commonly used for predicting continuous output. The working principle behind SVRs to find patterns in the data is to develop a suitable function like a hyperplane in a high-dimensional space that best fits the training data with a specified error threshold (ϵ) [58]. Traditional regression models like linear and polynomial

regression minimize the sum of squared errors between predicted and actual values; SVR aims to ensure that the error is within the specified threshold while keeping the model as flat as possible. The mathematical formulation of a support vector regressor is as follows:

$$\{(x_i, y_i)\}_{i=1}^n$$

$$f(x) = \omega_1 x_1 + \omega_2 x_2 + \dots \dots \omega_n x_n + b = \langle w, x \rangle + b$$

Where; x_i is the i th input feature vector, y_i is the corresponding target value, $f(x)$ is the function that deviates from the actual target values y_i , w and b are the weight and biases associated with the input feature vector, respectively. The model fitting using linear SVR can be explained as an optimization problem formulation i.e.

$$\min_{w, b, \xi, \xi^*} \frac{1}{2} \|w\|^2 + C \sum_{i=1}^n (\xi_i + \xi_i^*)$$

$$y_i - \langle w, x_i \rangle - b \leq \epsilon + \xi_i$$

$$\langle w, x_i \rangle + b - y_i \leq \epsilon + \xi_i^*$$

$$\xi_i, \xi_i^* \geq 0$$

Where; ξ_i and ξ_i^* are the slack variables introduced in the objective function to accommodate those data points that do not fit within the defined margin of error tolerance, ϵ . C in the objective function represents the regularization parameter that controls the trade-off between the flatness of the function and the amount up to which deviations larger than ϵ are tolerated. SVR uses the kernel function to efficiently capture the non-linear relationship between the input and output variables. Effective implementation of SVR involves selecting a suitable kernel function, regularization factor, and error tolerance [59]. Various hyperparameters can be adjusted to optimize the model predictions.

2.3.4 Gaussian Process Regression (GPR)

Unlike traditional regression methods, GPR is a non-parametric algorithm that uses a Bayesian statistical approach for regression that can be particularly effective for capturing complex non-linear relationships between input-output data pairs [60], [61]. It uses a covariance or kernel function to define the similarity between any two input space points. This kernel-based approach allows GPR to find a continuous distribution over the function that fits the data. It provides not only predictions but also estimates uncertainty in the predictions. The mathematical representation of GPR is as follows:

$$\{(x_i, y_i)\}_{i=1}^n$$

Where, $x_i \in R^d$ is a d-dimensional vector and y_i is the corresponding target value. The GPR relates the input and output variables with mean function μ and covariance function k and the distribution of function at a set of test points x^* can be given by:

$$f(x^*) \sim \mathcal{N}(\mu(x^*), k(x^*, x^*))$$

The mean and covariance functions are defined using the kernel function. The most commonly used kernel function is the squared exponential function, which can be defined as:

$$k(x_i, x_j) = \exp\left(-\frac{1}{2}\|x_i - x_j\|^2\right)$$

Based on the training data points, the GPR model uses Bayesian inference to learn the distribution of the function that is most likely to have generated the data, and it involves the computation of the posterior distribution of function, which is defined as:

$$p(f | x, y) = \frac{p(y | x, f)p(f)}{p(y | x)}$$

Where;

$y \mid x, f$ = Likelihood of the data given the function f

$p(f)$ = Distribution of function

$(y \mid x, f)$ = Marginal likelihood of the data.

After the posterior distribution of function has been learned, the model can make predictions on the test points x^* by calculating the posterior predictive distribution, which can be given as follows:

$$p(f^* \mid x^*, y, x) = \int p(f^* \mid x^*, f) p(f \mid y, x) df$$

The GPR works best for small datasets and gives predictions with uncertainty quantification. However, its effectiveness is reduced with large datasets due to the limitation of scalability. Moreover, GPR struggles with sparsity in both input features and prediction samples, posing limitations mainly when dealing with high-dimensional datasets [60].

2.3.5 Decision Tree Regression

A decision tree is also a non-parametric algorithm that predicts the output based on the decision rules concluded from the input data features [62]. It learns non-linear patterns within the data by a method called binary recursive partitioning, which includes splitting data into two partitions, commonly called branches, through an iterative process. The main aim of the splitting is to reduce the variance or bias within each division to ultimately arrive at a more accurate prediction [58], [59]. At the start, all the training data is grouped into a common node, known as the root node; from here, the algorithm starts the splitting of data in an optimal way by minimizing the error metrics such as mean-squared-error (MSE) or mean absolute error (MAE) for regression problems. The splitting rule is applied to all the new divisions, and this process is continuously repeated until each node reaches a specified node size, and it is then designated as a leaf node **Fig.13**.

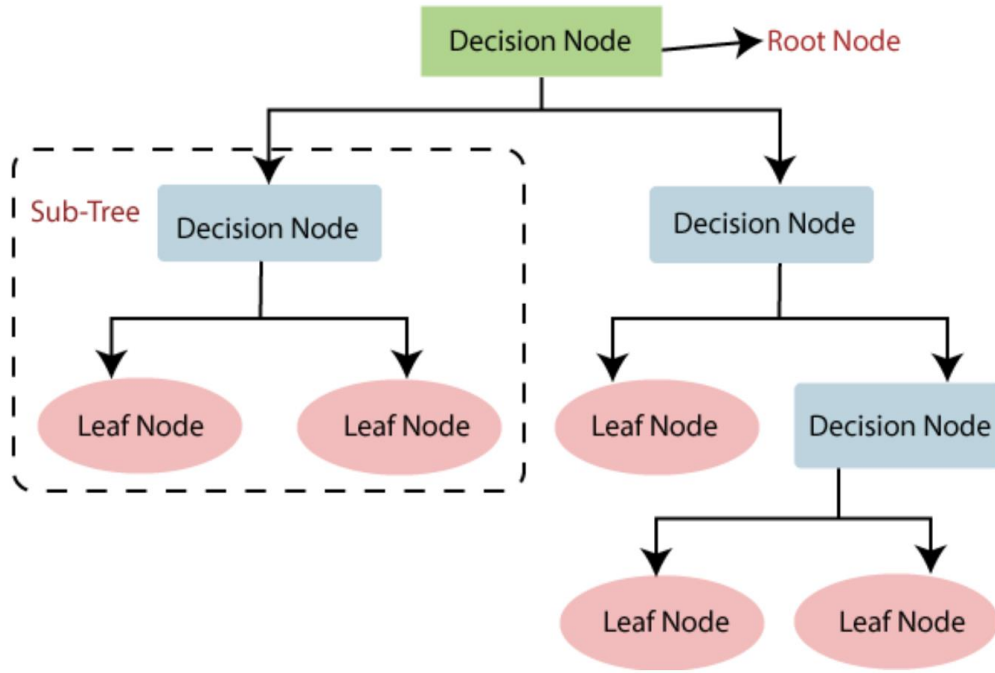


Figure 13: Decision tree pattern learning and key terminologies

2.3.6 Random Forest (RF) Regression

RF regression is an ensemble-based learning method that works by developing multiple decision trees during training and returns the average prediction of the individual trees to improve predictive accuracy [58]. The process of splitting in each decision tree within a RF stop under conditions similar to those in a single decision tree regression model. Additionally, RF applies bootstrapping (sampling with replacement) to generate diverse trees and uses random subsets of features when splitting nodes to increase variance among the trees, which enhances the model's generalization capability [59]. The aggregation of predictions across multiple trees reduces the risk of overfitting associated with single decision trees and leads to improved accuracy on unseen data. The algorithm's splitting process within individual trees ceases when it meets any of the predefined stopping criteria, ensuring that each tree maintains a balance between underfitting and overfitting. The mathematical representation of RF model is as follows:

$$f(x) = \frac{1}{n} \sum_{i=1}^n T(x)$$

Where $f(x)$ is the average prediction output from all the trees, n is the number of trees and $T(x)$ represents the output from a single tree.

2.3.7 Extreme Gradient Boosting (Xg-Boost) Regression

Xg-Boost is an efficient implementation of the gradient boosting framework that has become a popular choice for regression problems among many data scientists and ML practitioners. The strength of Xg-Boost lies in its ability to create a highly effective ensemble of decision trees designed to predict continuous outcomes, making it a robust choice for solving complex regression problems. The Xg-Boost regression model operates by sequentially adding decision trees, where each subsequent tree attempts to correct the errors made by the previously built trees [59]. This process continuously operates iteratively, adding trees until a specified stopping criterion is met. It is very important to mention that the gradient boosting algorithm aggregates the results of all the decision trees along the way to report the model performance, while the results of decision trees are calculated at the end of each process. The mathematical representation of the Xg-Boost algorithm includes the minimization of the objective function comprised of the loss function and the regularization term, as shown below:

$$f^{(t)} = \sum_{i=1}^N L(y_i, \hat{y}_i) + \sum_{k=1}^K \Omega(f_k)$$

Where $f^{(t)}$ is the objective function, $\Omega(f_k)$ is the regularization term, and $L(y_i, \hat{y}_i)$ is the training loss.

2.3.8 Artificial Neural Network

Artificial Neural Networks (ANNs) have emerged as one of the most robust tools in the field of ML because of their ability to capture the complex non-linear relations between data points

[63]. ANN models are inspired by the working principle of the human brain's biological neural networks and consist of interconnected artificial neurons called nodes. These nodes work together to process and transmit information, mapping the non-linear and complex data relationship. The structure of an ANN comprises an input layer, which has several nodes corresponding to the total input features, and an output layer, where each node represents a target variable. The hidden layers between input and output are crucial in ANN model architecture and comprise a number of adjustable nodes/neurons to avoid overfitting and underfitting the model (**Fig.14**) [64].

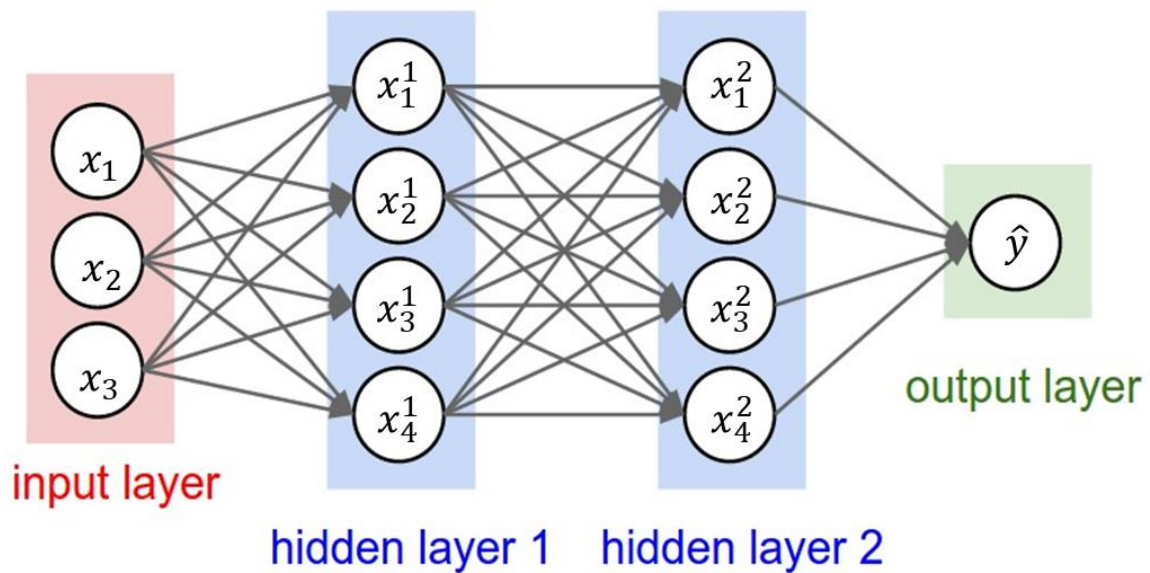


Figure 14: Artificial Neural Network architecture

The function of each neuron in the hidden layer is to take information in the form of an array from the previous layer, multiply it with the weight factor, add the bias term to their weighted sum, and pass it to the next layer (**Fig.15**).

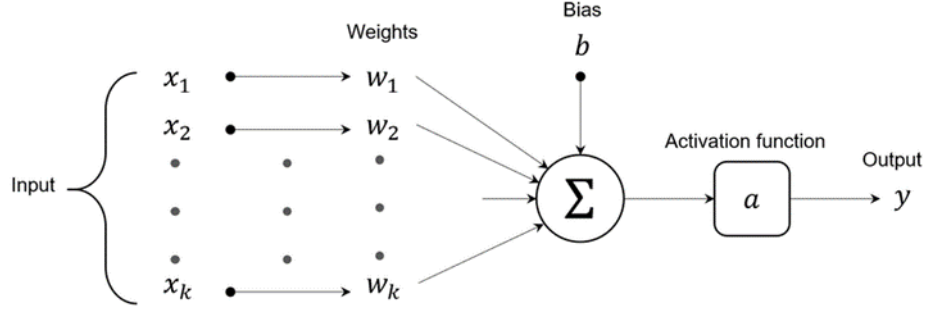


Figure 15: Operations within each neuron of Artificial Neural Network

Each neuron in the hidden layer does a computation based on the selected activation function. The objective of an activation function in neurons/nodes is to introduce non-linearity to enable complex data pattern recognition. Common types of activation functions include Sigmoid used for binary classification, Hyperbolic Tangent ($\tanh$) for centered data, and Rectified Linear Unit (ReLU) for enhancing deep learning efficiency and solving vanishing gradients issues [64]. The choice of activation function plays a crucial role in training efficiency and task-specific performance of the model.

During the learning phase within ANN model, the input variables are sequentially passed from the input layer to the hidden layer and finally to the output layer to predict the target variable. During the model's training phase, the value predicted by the network is compared with the actual values. The main goal is to minimize the error between these two values. This is achieved through a method known as backpropagation [64]. In backpropagation, the network adjusts its internal parameters (weights and biases) based on the error observed between the predicted and actual values. This adjustment is crucial in refining the model's accuracy and enabling it to make more precise predictions as the training progresses.

A comprehensive comparison of different surrogate modeling methods is shown in **Table.2**.

Table 2: Characteristics of different surrogate modeling methods

Characteristic	ML-Based Surrogate Models	Response Surface Methodology (RSM)	Taguchi Method
Computational Time	High (due to the extensive data processing)	Moderate (depends on the complexity of the design and analysis)	Low to Moderate (due to limited experimental design)
Accuracy	High (with sufficient data and complex models, can achieve high accuracy even in nonlinear and high-dimensional spaces)	Moderate to High (depends on the fit of the polynomial model to the data, more suited for quadratic relationships)	Moderate (focuses on robustness rather than pinpoint accuracy; optimal settings are derived to minimize variation)
Handling Non-Linearity	Excellent (capable of modeling highly non-linear relationships)	Good (can handle non-linearity to a certain extent with polynomial models)	Low to Moderate (primarily focused on identifying robust settings rather than modeling the underlying process)
Dimensionality	Excellent (can handle high-dimensional data effectively, although it may require more data and computational power)	Moderate (effectiveness decreases as the number of variables increases)	Low to Moderate (suitable for a smaller number of factors due to the orthogonal array design constraints)
Applying Complexity	High (suitable for complex problems with large datasets, capable of capturing complex relationships)	Moderate (effective for moderately complex problems, where a quadratic approximation is sufficient)	Low (best applied to simpler problems where the goal is to reduce variability and improve robustness)
Design of Experiments (DoE)	Space-filling designs (e.g., LHS, SOBOLE, MC)	Factorial designs, including Central Composite Design (CCD) and Box-Behnken Design	Orthogonal arrays for efficient, structured experimentation.
Data Requirement	High (requires large datasets to capture complex relationships and ensure model robustness effectively)	Moderate (needs enough data points to fit the model but generally fewer than ML-based models)	Low to Moderate (designed to work efficiently with a smaller number of experiments)

2.3.9 Literature review of ML-based surrogate modeling

ML-based surrogate modeling has gained significant importance as it can handle complex data and is able to efficiently capture the non-linear relations between input-output variables [13], [14]. ML-based models can also generalize the new, unseen data, making it an excellent choice for predicting and optimizing real-world scenarios. Many different applications of surrogate modeling in chemical engineering have already been explored. Some of the previous work related to surrogate modeling of industrial processes and simulation-based optimization are as follows:

Ismail et al. [65] have developed a superstructure optimization model to identify the optimum configuration for the biodiesel production plant to minimize the net present cost of the final product. Artificial neural network (ANNs) is deployed to construct the surrogate model to replace the unit operations, thermodynamics, and mixing models to reduce the computational load. The results show that the prepared surrogate models notably reduced the computation cost and time. Hough et al. [66] have developed a kinetic model using ANN and a decision tree for the biogas pyrolysis reacting system. The main goal was to reduce the computational load and time. The formulated model can predict output from the reacting system with 99.9% accuracy. Chew et al. [67] have developed an ANN and random forest-based predictive model for the circulating fluidized bed reactor. The main goal was to predict the output from the reactor with the variation in the input operating conditions without using the first principal methods and to reduce the computational expenses. The input data is generated through the Python SALib package based on a quasi-random Sobol sequence that efficiently covers the input parameter space. The trained model shows promising results and can predict the reactor input-output relation with 99.9% accuracy. Manhee et al. [68] studied the environmental impact and techno-economic feasibility study of hydrogen production from the methanol steam reforming (MSR) process in a series of membrane reactors. Twelve different input parameters that can directly

impact the techno-economic of the process are selected. A ML-based ANN and support vector predictive model is established, having an output of total carbon emission, hydrogen production cost, and hydrogen production rate. It is concluded that the number of reactors and operating conditions, mainly the reactor temperature, are the key parameters that affect the system's overall economics. Ladan et al. [69] have combined the Design of the Experiment (DoE) with ML to develop a ML model that can predict the energy consumption and purity of the final product. Techniques like Principal component analysis (PCA), standardization, and normalization are applied to reduce and clean the dataset for efficient implementation of ML models. The training data is generated using the three-level DoE technique. Radial basis functional neural network (RBF-NN) is used for the model training, and it is evaluated on standard statistical performance metrics like R^2 , root mean square error, and average absolute relative deviation (%AARD). The proposed ML model can predict energy consumption and final product purity of a typical gas processing plant with a confidence level of 98.6%. Xiaoti et al. [70] have studied surrogate model implementation for the methanol production plant under dynamic operation conditions. Training data from process simulation software like Aspen is generated under dynamic conditions with a ramp rate of 50% per hour. ML dynamic type non-linear autoregressive exogenous model (NARX) is deployed to model the input-output behavior of the process. The trained NARX model exhibited good accuracy with the $MAE = 2.30 \times 10^{-4}$, $R^2 = 1.0$, and $RMSE = 3.09 \times 10^{-5}$. Kaiwen et al. [14] studied the energy cost optimization of the extractive distillation process using the Aspen simulator. Two approaches of surrogate modeling are used: one is the algebraic model, and the second one is the neural network model with rectifier (ReLU). The trained models' performance is compared based on accuracy and the results reveals that the algebraic model performs better in less complex systems than the neural network model, which performs better in complex systems.

2.4 Design of Experiments

The reliability and performance of surrogate models significantly depend on the data quality used for the training. Design of Experiments (DoEs) is the commonly used strategy to prepare good-quality training datasets. This strategy helps to generate sample points systemically. It covers the effects and variation of all the input variables along the specified range of the target variables involved in the system of interest [55]. DoEs strategy is followed by the step to generate data from the high-fidelity simulation model. The extent of information that can be obtained from the simulation models depends on the training data size, while the computational constraints often limit the number of samples that can be generated [71]. For surrogate modeling, a sample size of 10-times that of the number of variables under consideration is recommended to effectively capture the complex relations between the input-output variables [13]. In this section, two different DoE strategies are explained for generating datasets i.e. Factorial designs and space-filling designs. The relevant concepts of these DoEs strategies are as follows:

2.4.1 Factorial design:

Factorial design is an efficient strategy used for designing experiments that aim to analyze the effects of multiple variables across different levels on one or more response variables of interest [72]. In a factorial design, “factor” is any variable that significantly affects the experimental results, while the “level” is the specific value of a factor within the range. For example, both temperature and pressure are the key factors of a reactor that can affect the conversion, and a three-level design for temperature could be 20°C, 30°C, and 40°C. Factorial DoEs allow for estimating the main effects of design variables without requiring extensive experimental runs [71], [72]. The factorial design sample points are typically the design space's corners, edges, and face points. There are different Factorial designs depending on the number of factors

involved, the level of each factor, and the need to control the experimental error or variability.

Some of the Factorial designs are demonstrated in the **Fig.16**.

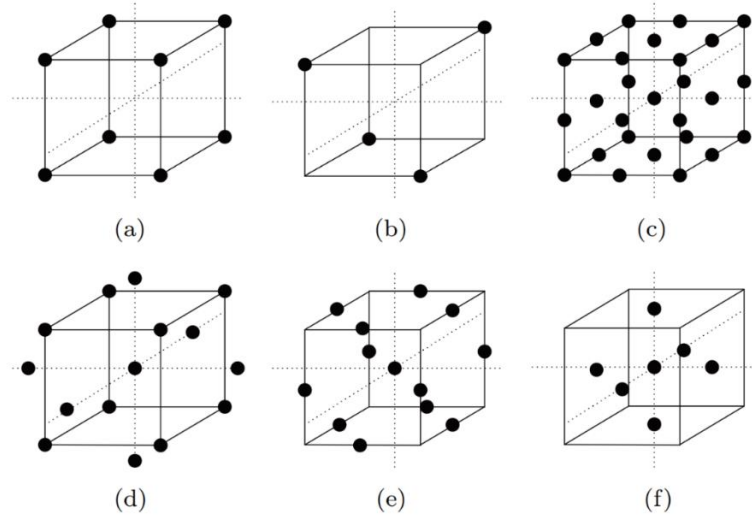


Figure 16: Factorial design examples: (a) Full factorial design, (b) Fractional factorial design, (c) Block design, (d) Central composite design, (e) box-Behnken design, and (f) star distribution. [71]

2.4.2 Space-filling experimental designs

Space-filling designs are the strategical methodologies for designing experimental sample points in a multidimensional space. The data for developing a surrogate model comes from the simulation; therefore, the chosen points for the experiments are desired to cover the entire multi-dimensional space of variables uniformly. This approach is commonly used in designing computer-based simulations and global optimization, where developing the behavior of a process across a wide range of variables is crucial [71]. Following are the key characteristics of a space-filling design.

- **Uniform Coverage:** Space-filling design distributes the experimental points uniformly and ensures that all regions of the multi-dimensional space are explored.
- **Minimized Clustering:** It ensures that each experimental point is designed to minimize the clustering to avoid oversampling from certain areas.

- **Maximized Minimum Distance:** Some space filling designs tend to maximize the minimum distance between any two experimental points to avoid the likelihood of points being too close to each other.

The most commonly used space-filling designs are Monte Carlo Sampling (MCS), SOBOL sequence sampling, and Latin hypercube sampling (LHS). MCS is a straightforward random sampling technique that produces a sample point from the probability distribution of each variable under the observation. **Fig.17** provides an illustrative example of ten sample points generated using MC sampling in the space of two variables. This method is not stratified, and it creates sampling space with uneven distribution of the variables; some regions may be over-represented while others may be under-represented [73].

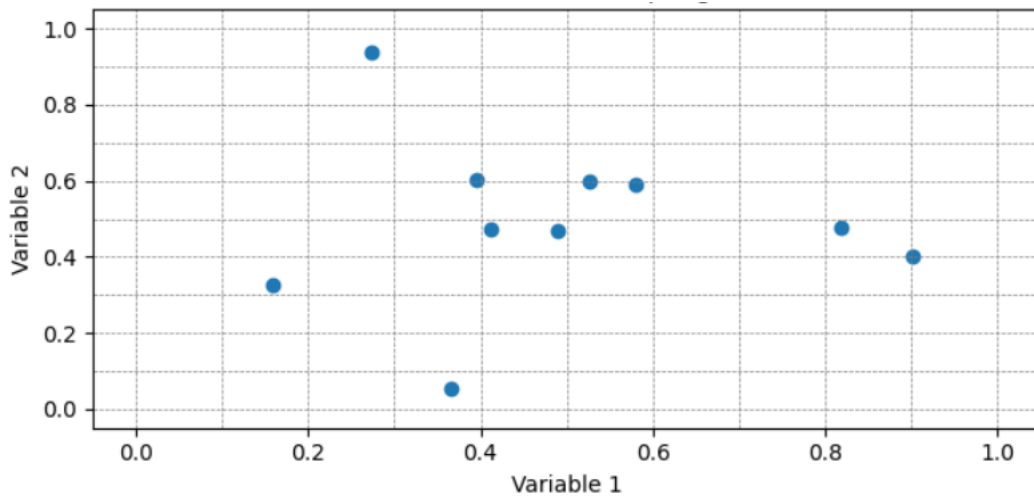


Figure 17: Illustrative MCS design of ten experimental points in two-dimensional variable space

Sobel sequence sampling is quasi-random, which means it provides a good balance between randomness and uniform coverage of the variable under observation but requires more computational understanding, especially when dealing with datasets with higher dimensions. **Fig.18** illustrates ten sample points generated using SOBOL sequence sampling in two-dimensional variable space.

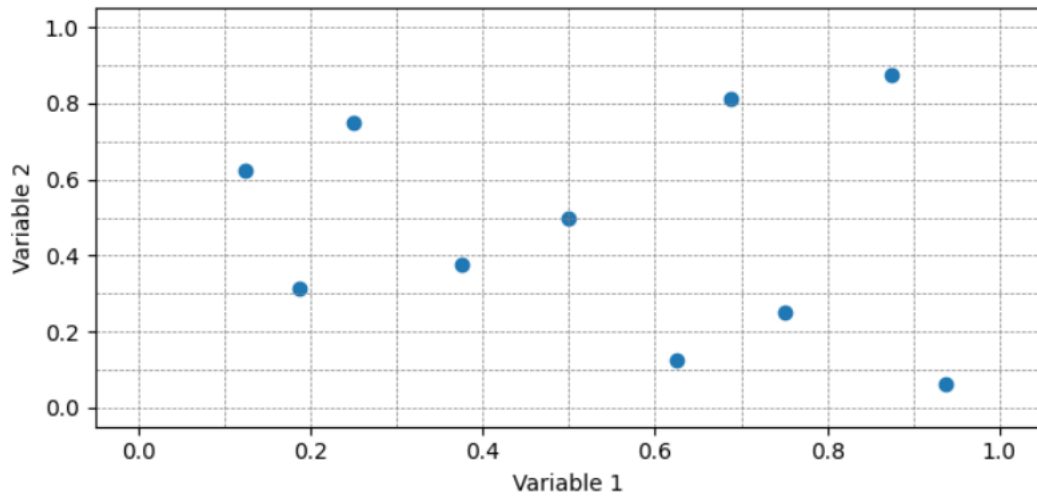


Figure 18: Illustrative Sobol sequence design of ten experimental points in two-dimensional variable space

On the other hand, Latin Hypercube sampling is a stratified sampling technique that ensures each sample point is the only unique one in its "row" and "column" in a multi-dimensional space [73]. **Fig.19** illustrates the LHS space-filling design of ten sample points in two-dimensional variable space.

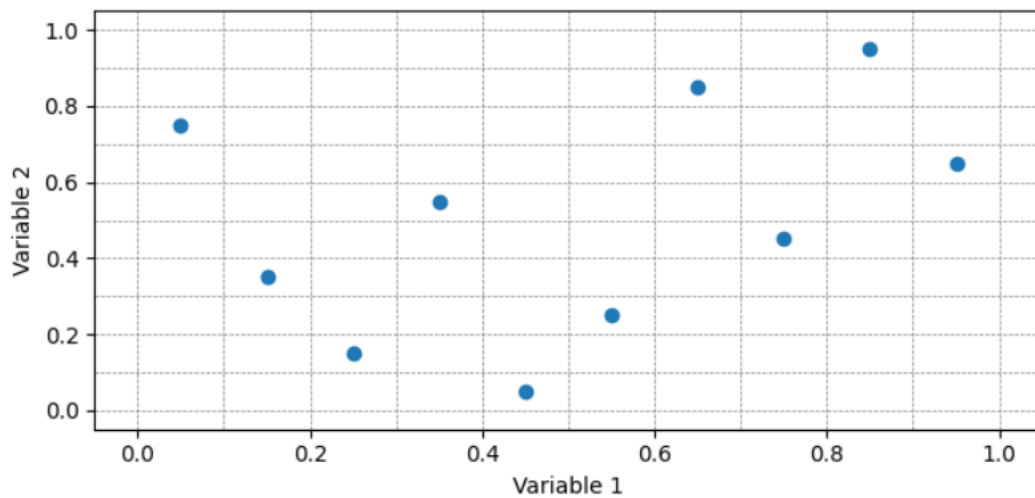


Figure 19: Illustrative LHS space-filling design of ten experimental points in two-dimensional variable space

LHS sampling can be more efficient than other random sampling methods as it covers the sampling space more evenly. There are different variants of LHS space-filling design, and **Fig.20** provides an illustration of different criteria available for creating LHS space-filling designs.

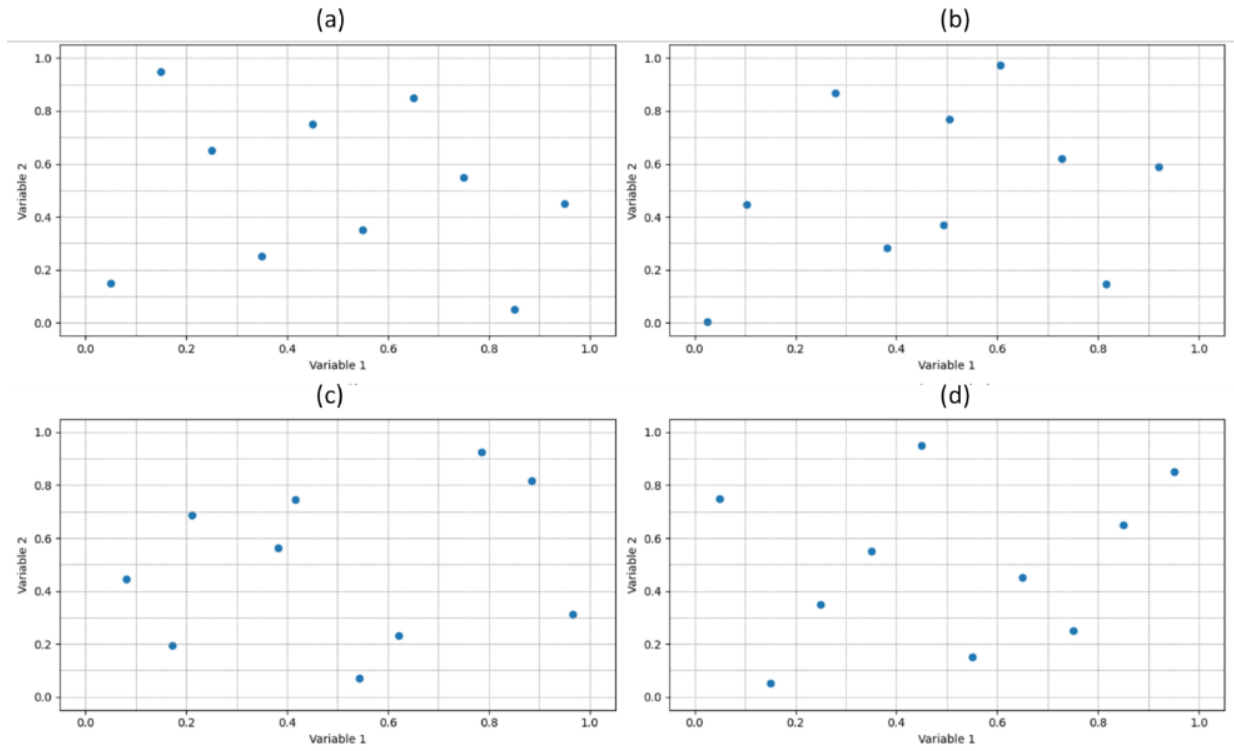


Figure 20: Different criteria for LHS space-filling design (a) Center (b) Max-min (c) Correlations (d) Center max-min

There is no universal rule for the selection of the sampling technique, and increasing the sample space will not always guarantee better model accuracy, so experimentation with different methods is required [74].

Chapter 3. Research and Methodology

This chapter covers the research methodology adopted in this work. This includes the description of the methanol production plant under consideration and the specific processing unit intended for modeling. Data generation methodology, the ML modeling method adopted in this research, and the evaluation metrics selected for testing performance are discussed. Furthermore, the multi-objective optimization method is discussed. A comprehensive overall methodology of this research is shown in **Fig.21**.

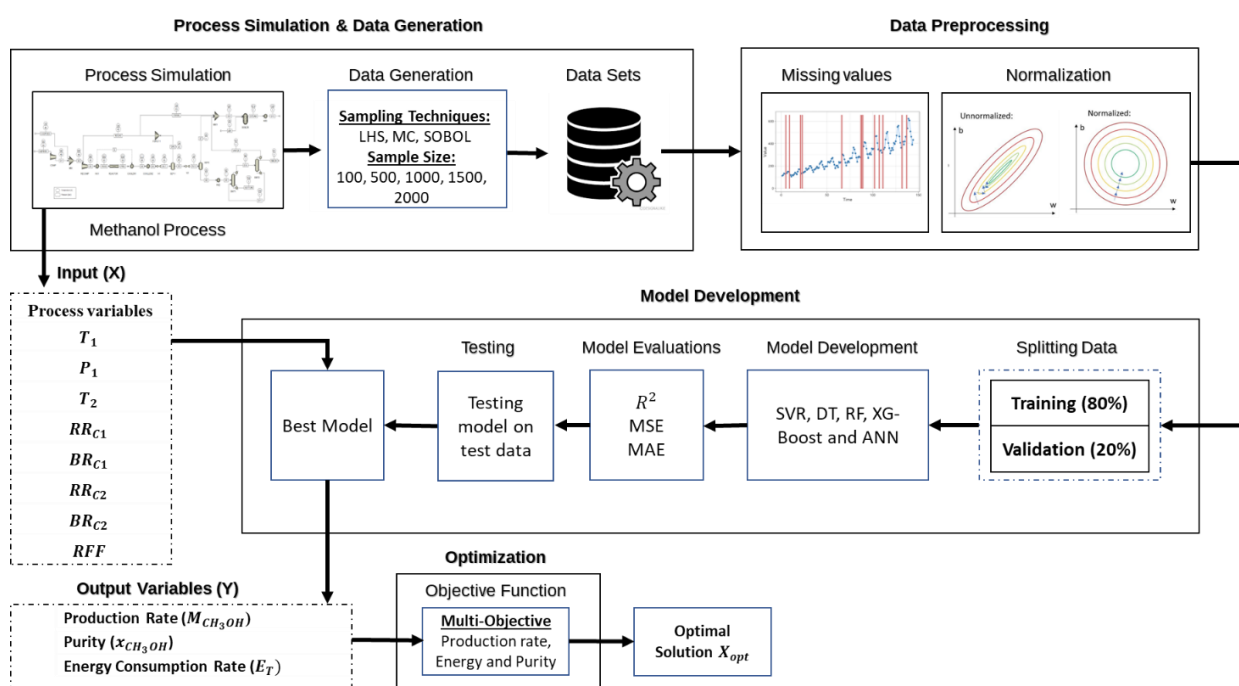


Figure 21: Flow chart of overall methodology (model development and optimization)

3.1 Process simulation of methanol production plant

Renewable methanol is produced by the catalytic hydrogenation of carbon dioxide derived from the industrial capturing system, and green hydrogen is produced by electrolysis. Methanol production is an exothermic process and takes place in the presence of the Cu/ZnO/ Al₂O₃ catalyst. The reaction operating conditions range from 20 to 100 bar and 200 – 300°C [70]. A continuous methanol production process is adopted from Gu et al.'s [75] work and is modeled

on the Aspen plus V.14. simulator. The original work was based on the Graff kinetic model, and it is replaced with the Vanden Bussche–Froment due to its accurate representation of the reaction mechanisms and kinetics as compared to the Graff kinetic model, which lacks incorporating the variation in the variable process conditions [26]. The details of the reaction mechanism and kinetic parameters are provided in **Table 3-6**.

Table 3: Rate equations for Graaf and Vanden Bussche-Froment kinetic models

Kinetics type	Rate Equations (mol/kg. s)
Graff Kinetic model [76]	$r_{\text{CO}_2 \rightarrow \text{MeOH}} = \frac{k_1 K_{\text{CO}_2} \cdot f_{\text{CO}_2} f_{\text{H}_2}^{1.5} - \frac{k_1 K_{\text{CO}_2}}{K_{\text{eqCO}}} \cdot f_{\text{MeOH}} f_{\text{H}_2\text{O}} f_{\text{H}_2}^{-1.5}}{1 + K_{\text{CO}} \cdot f_{\text{CO}} + K_{\text{CO}_2} \cdot f_{\text{CO}_2} f_{\text{H}_2}^{0.5} + K_{\text{CO}_2} \cdot \frac{K_{\text{H}_2\text{O}}}{K_{\text{H}_2}^{0.5}} \cdot f_{\text{CO}_2} f_{\text{H}_2\text{O}}}$ $r_{\text{RWGS}} = \frac{k_2 K_{\text{CO}_2} \cdot f_{\text{CO}_2} f_{\text{H}_2} - \frac{k_2 K_{\text{CO}_2}}{K_{\text{eqRWGS}}} \cdot f_{\text{H}_2\text{O}} f_{\text{CO}}}{1 + K_{\text{CO}} \cdot f_{\text{CO}} + K_{\text{CO}_2} \cdot f_{\text{CO}_2} f_{\text{H}_2}^{0.5} + K_{\text{CO}_2} \cdot \frac{K_{\text{H}_2\text{O}}}{K_{\text{H}_2}^{0.5}} \cdot f_{\text{CO}_2} f_{\text{H}_2\text{O}}}$ $r_{\text{CO} \rightarrow \text{MeOH}} = \frac{k_3 K_{\text{CO}} \cdot f_{\text{CO}_2} f_{\text{H}_2}^{1.5} - \frac{k_3 K_{\text{CO}}}{K_{\text{eqCO}}} \cdot f_{\text{MeOH}} f_{\text{H}_2}^{-0.5}}{1 + K_{\text{CO}} \cdot f_{\text{CO}} + K_{\text{CO}_2} \cdot f_{\text{CO}_2} f_{\text{H}_2}^{0.5} + K_{\text{CO}_2} \cdot \frac{K_{\text{H}_2\text{O}}}{K_{\text{H}_2}^{0.5}} \cdot f_{\text{CO}_2} f_{\text{H}_2\text{O}}}$
Vanden Bussche–Froment Kinetic model [25]	$r_{\text{CO}_2 \rightarrow \text{MeOH}} = \frac{k_1 p_{\text{CO}_2} p_{\text{H}_2} - \frac{k_1}{K_{\text{eqCO}_2}} p_{\text{MeOH}} p_{\text{H}_2\text{O}} p_{\text{H}_2}^{-2}}{\left(1 + K_1 \frac{p_{\text{H}_2\text{O}}}{p_{\text{H}_2}} + \sqrt{K_2} \cdot p_{\text{H}_2}^{0.5} + K_3 p_{\text{H}_2\text{O}}\right)^3}$ $r_{\text{RWGS}} = \frac{k_2 p_{\text{CO}_2} - \frac{k_2}{K_{\text{eqRWGS}}} p_{\text{CO}} p_{\text{H}_2\text{O}} p_{\text{H}_2}^{-1}}{1 + K_1 \frac{p_{\text{H}_2\text{O}}}{p_{\text{H}_2}} + \sqrt{K_2} \cdot p_{\text{H}_2}^{0.5} + K_3 p_{\text{H}_2\text{O}}}$

Table 4: Kinetic, Adsorption and Equilibrium expression for VBF kinetic models [25], [26]

Parameters	Vanden Bussche–Froment Kinetics
Kinetic parameters	$k_1 = 1.07 \times 10^{-13} e^{(36696/RT)}$ $k_2 = 1.22 \times 10^2 e^{(-94765/RT)}$

Adsorption Coefficient	$K_1 = 3.45 \times 10^3$ $\sqrt{K_2} = 4.99 \times 10^{-1} e^{17197/RT}$ $K_3 = 6.62 \times 10^{-11} e^{124119/RT}$
Equilibrium Coefficient	$\log_{10} (K_{eq,CO_2}) = \frac{3066}{T} - 10.592$ $\log_{10} (K_{eq,RWGS}) = -\frac{2073}{T} + 2.029$

Table 5: Exponential and pre-exponential factors for VBF kinetic models [25], [26]

Reaction	Kinetic Parameters	Vanden Bussche–Froment Kinetics	
		A	E _{act}
CO ₂ hydrogenation	k_{dir}	1.07×10^{-13}	3.67×10^4
	k_{rev}	4.18×10^7	2.20×10^4
RWGS	k_{dir}	1.22×10^2	9.48×10^4
	k_{rev}	1.1412	5.51×10^4

Table 6: Adsorption coefficient of the VBF kinetic models [25], [26]

Adsorption Constants	Vanden Bussche–Froment Kinetics	
	B	E _{ads}
K_1	3.45×10^3	-
$\sqrt{K_2}$	4.99×10^{-1}	-1.72×10^4
K_3	6.62×10^{-11}	-1.24×10^5

The kinetic parameters and the adsorption coefficients can be defined using the Arrhenius equation:

$$k_i = A_i \cdot \exp \left(-\frac{E_j}{RT} \right)$$

$$K_i = B_i \exp \left(-\frac{E_i}{RT} \right)$$

The Aspen acceptable format for kinetics and adsorption factors can be defined as follows:

$$\ln k_i = A_i - \frac{E_j}{RT}$$

$$\ln K_i = B_i - \frac{E_j}{RT}$$

The kinetic and adsorption parameters are converted to Aspen acceptable format for the reactor modeling within in Aspen environment. For the methanol production process, the CO₂ can be assumed to be captured from the steel, cement, or coal-based power plants reaching the methanol production site at a pressure of 40 bar. The composition of the captured carbon dioxide stream depends on the specific technology used and the processes involved. For simplicity pure carbon is considered as post-combustion capturing technologies can achieve up to 99% purity [77]. The hydrogen produced from the PEM electrolyzer can reach up to 99.99% purity [78] plant site at 30 bar pressure, which needs to be immediately used for the methanol production process. The detailed Aspen Plus flow sheet for the methanol production process is shown in **Fig.22**. Soave-Redlich-Kwang and peng-robinson property methods are used for the simulation of the methanol production process as recommended in the literature [75]. The carbon dioxide (14 °C, 40 bar) and hydrogen (50 °C, 30 bar) streams are mixed in the ratio of 1:3. They are combined with the recycle stream and fed to the packed bed tubular reactor operated at 75 bar and 230 °C. The reactor products are cooled down to 30 °C, and most unreacted gaseous products are recycled back to the reactor. The liquid stream passes through a distillation section where methanol is purified in a two-stage distillation process. The overall methanol production process is shown in **Fig.22**. The simulation's key parameters are given in **Table.7**.

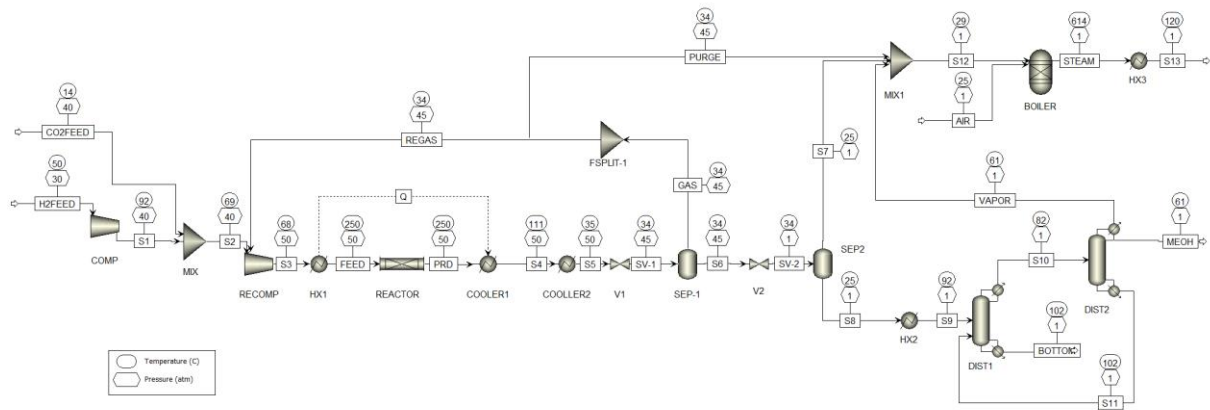


Figure 22: Aspen plus simulation of the methanol production process [75]

Table 7: Methanol Simulation Process Parameters

Unit	Item	Value
Reactor	Number of tubes	81
	Reactor Length (meter)	12
	Diameter (meter)	0.06
	Catalyst loading in reactor(kg)	86.5
	Bed voidage fraction	0.55
DIST1	Reflux Ratio	1.11
	Feed Stage	15
	Number of Stages	30
	Boil-up ratio	0.6
	Type of Reboiler	Kettle-type
	Type of Condenser	Partial-Vapor
	Operating Pressure	1 atm
DIST2	Reflux Ratio	1.11
	Feed Stage	15
	Number of Stages	30
	Boil-up ratio	0.8
	Type of Reboiler	Kettle-type
	Type of Condenser	Partial-Vapor
	Operating Pressure	1 atm

3.2 Data generation

Once the simulation is functional, the next step is to generate good good-quality dataset that covers all the variations of the selected variables for the data-driven model development. The key parameters that significantly impact the energy requirement, methanol production rate, and purity are identified, and the operating range is defined to study the impact of their variation on the target variables. The choice of input and output parameters that are selected for this study are listed in **Table.8**.

Table 8: Input and output pair selected for the surrogate modeling of methanol production process

	Variables	Ranges	Description
Input Variables	T_1	200-300 (°C)	Reactor operating temperature
	P_1	50-100 (bar)	Reactor operating pressure
	T_2	85-95 (°C)	Feed Inlet temperature to the distillation column
	RR_{C1}	1-15	Reflux ratio in first distillation column
	BR_{C1}	0.8-15	Boil up ratio in first distillation column
	RR_{C2}	1-15	Reflux ratio in second distillation column
	BR_{C2}	0.6-15	Boil up ratio in second distillation column
	RFF	0.5 – 0.99	Recycle flow fraction to the reactor
Output Variables	M_{CH_3OH}	kg/hr	Mass flow rate of methanol product
	x_{CH_3OH}	-	Mass fraction of methanol in final product (Purity)
	E_T	kW	Energy consumption rate

To generate good quality dataset there are different sampling techniques, each works on a different criterion, as discussed in section 2.4. This study adopts a comparable approach, employing three distinct techniques—Latin Hyper Cube, Monte Carlo, and Sobel—to generate sample points of varying sizes (100, 500, 1000, 1500, and 2000). The objective is to comprehensively investigate the influence of both sampling technique and sample size on the performance of the ML model. The sample data points for LHS and SOBOL are generated in the Python environment using “pyDOE-2” and “sobol_seq” libraries, respectively, and for Monte Carlo random sampling points, a Python code is utilized, developed using "numpy"

library. **Table.9** summarizes the data generated using simulation using LHS, Monte Carlo, and Sobel with varying sample sizes. The number of feasible simulations along with training and test data split is also presented.

Table 9: Data set generated from the simulation using different sampling technique and sample size

Sampling technique	Sampling Size	Feasible simulations	Training set	Validation set
Latin Hyper Cube	100	94	75	19
	500	485	388	97
	1000	970	776	194
	1500	1453	1162	291
	2000	1931	1544	387
Monte Carlo	100	98	78	20
	500	481	384	97
	1000	965	772	193
	1500	1457	1165	292
	2000	1924	1539	385
SOBOL	100	98	78	20
	500	478	382	96
	1000	958	766	192
	1500	1438	1150	288
	2000	1923	1538	385

3.3 Data preprocessing:

Data preprocessing is a crucial step in ML model development, which involves preparing and cleaning the raw data for training an ML model. This step can impact the performance and accuracy of the ML model. It involves splitting the data into training, validation, and test split, removing outliers or missing values, and normalization and standardization of data. For this study, we selected the 80% and 20% data split for training and validation. A separate test data set of 200 sample points is also generated, serving as a completely unseen dataset for the final model testing. In our dataset, some un-converged sample points need to be removed; otherwise, they can affect the model's training and performance. It is also a good practice to normalize or

standardize the data before training a model to ensure that all the features or variables are on the same scale to prevent the disproportionality resulting from the different feature magnitudes. Normalization adjusts the range of all the variables involved in the system to a standard scale between 0 and 1, while standardization transforms the data to have a mean of 0 and a standard deviation of 1. Implementing these techniques makes the model more robust and less sensitive to the variation in the variable's magnitude, which ultimately enhances the convergence speed and overall model performance during training. The scikit-learn built-in train-test split and MinMax normalization function are utilized for effective data preprocessing. The formula for normalizing the dataset is as follows:

$$x_{\text{normalized}} = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

Where; $x_{\min}$ and $x_{\max}$ represent minimum and maximum values of the variables in the data set, and x represent the actual value that needs to be normalized in the dataset.

3.4 Model Training

In this study, five machine learning models are trained on the dataset generated using different sampling techniques. First, the data set mentioned in Table.9 is trained using SVR model to predict the methanol production rate. The purpose of training the model on a dataset generated using different sampling techniques and sample sizes is to analyze their effect on model performance. The best dataset is chosen based on model performance and is used to screen different ML models. This study uses four machine learning methods, i.e., SVR, DT, RF, and Xg-boost, as well as a DL artificial neural network for performance comparison. The dataset is loaded into the Python environment using pandas in Microsoft Visual Studio Code software. After that, the data will be preprocessed. It involves splitting the data into training, validation, and test split, removing outliers or missing values, and normalization and standardization of data. For this study, an 80-20% training and validation set split is created using the

train_test_split function from scikit-learn library. The data is normalized using a min-max normalizer, and the unconverted sample points are removed from the dataset. First, models are trained on the training data set, and their performances are evaluated using the validation set. A separate test dataset of 200 sample points is utilized for final performance evaluation to report their accuracy. The SVR, DT, RF, and Xg-boost regressor models are implemented using scikit-learn and Xg-boost libraries. Tensorflow and Keras libraries are employed to develop a deep-learning artificial neural network model. The hyperparameters used for the training of different models are shown in **Table10**.

Table 10: Hyperparameters used for model training

Models	Adjustable Hyperparameters
"SVR"	"C" = 1.0, "kernel" = 'rbf', "Epsilon" = 0.1, "degree" = 3
"RF"	N_estimators = 100, Criterion: "mse", Max depth= None, Minimum sample split = 2
"DT"	Criterion: "mse", Splitter: "best", Max depth= None, Minimum sample split = 2
"Xg-Boost"	N_estimators = 100, Learning rate = 0.1, max_depth = 6, booster = "gbtree"
"ANN"	Hidden layers = 2, Number of neurons in each layer = 8, Activation function hidden-layer = ReLU, Activation function output-layer = Linear, Batch Size = 128

3.5 Model Evaluation

The performance of ML models is evaluated on the coefficient of determination (R^2), mean-squared error (MSE), and mean-absolute-error (MAE) performance metrics.

$$R^2 = 1 - \frac{\text{Sum of square of residual}}{\text{total sum of squares}}$$

$$MSE = \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{N}$$

$$MAE = \frac{\sum_{i=1}^N |y_i - \hat{y}_i|}{N}$$

Where; y_i , $\hat{y}_i$ are the actual and predicted values, and N represents the number of data points.

3.6 Optimization

For the optimization of process operating conditions, two algorithms are used in this work i.e., Sequential Least Square Quadratic Programming (SLSQP) and Differential Evolution Algorithm (DE) algorithm. The DE algorithm is a heuristic approach for global optimization based on the theory of natural selection and evolution and is similar to the Genetic Algorithm (GA) [79]. This algorithm is used to minimize the function and works on real values. Implementing both DE and genetic algorithms involves designing three operators i.e., selection, crossover, and mutation [80]. The only difference between them is the order in which these operators are implemented. The DE algorithm starts with an initial population of potential solutions (vectors) to the optimization problem and iteratively improves the results based on the rules of mutation, crossover, and selection. The key steps involved in the DE algorithm are as follows:

- **Initialization:** The first step is to randomly generate an initial population of potential solutions (vectors) from the bounds of the optimization problem.

- **Mutation:** The second step is to create mutant vector for each individual in the population. This process typically involves selecting three vectors (A, B, C) randomly from the population and creating a new vector by adding a weighted difference between two vectors (B and C) to the third vector (A). The formula for the mutation is as follows:

$$V_i = A + S \cdot (B - C)$$

Where ‘S’ is the scaling factor used for controlling the rate of mutation.

- **Crossover (Recombination):** In this step, the mutated vector is mixed with the current individual (vectors) to produce a trial vector. The mixing of vectors is controlled by a parameter called crossover rate that ensures the population's diversity. The elements of the trial vector are chosen from either the mutated vector or the target vector based on a random comparison with the crossover rate.
- **Selection:** The performance of the trial vector is evaluated, and its fitness is compared to that of the original vector in the population. If the produced trial vector has better fitness, it is replaced with the original vector in the population; otherwise, the original vector is retained.
- **Termination:** The previously mentioned steps are repeated over a whole generation until the termination criterion is met. It could be the number of generations, a threshold fitness value, or any other user-defined condition.

DE algorithm is an efficient algorithm that does not require the gradient of the function to be known, thus making it suitable for problems including non-differentiable functions and enabling it to solve complex and non-linear problems efficiently[80]. The limitation of DE algorithm includes slow convergence speed, especially problems with high-dimension and when the initial population is far from the optimum.

Chapter 4. Results and Discussions

4.1 Process Simulation

A continuous methanol production plant is modeled using the process and operating conditions explained in section 3.1. First, only the reactor model developed using Vanden Bussche–Froment (VBF) kinetics is validated with the literature values [81] to check the reliability. For validation, all the process conditions are selected to be exactly the same in the literature [81], i.e., feed composition, property method, reactor design, and operating conditions. The validation results of the reactor model are shown in Table.11. which reveals good agreement with the literature values with a small acceptable error. The sensitivity analysis of the reactor operating conditions on the selectivity of the final product is also compared with the literature [81] (Fig.23), which shows that at temperatures of 250°C and 150 bar pressure the selectivity of the methanol is the highest.

Table 11: Validation of reactor model with the literature values

Parameters	Input Conditions	Literature [81]Reactor Output results	Simulated Reactor Output results	% Error
Temperature (°C)	150	267	267	0%
Pressure (bar)	110	110	108.562	1.3%
CO (kmol/hr)	4066	1234.43	1232.47	0.15%
H ₂ (kmol/hr)	28920	20716.3	20713.2	0.014%
CO ₂ (kmol/hr)	3976	3129.42	3129.15	-0.008%

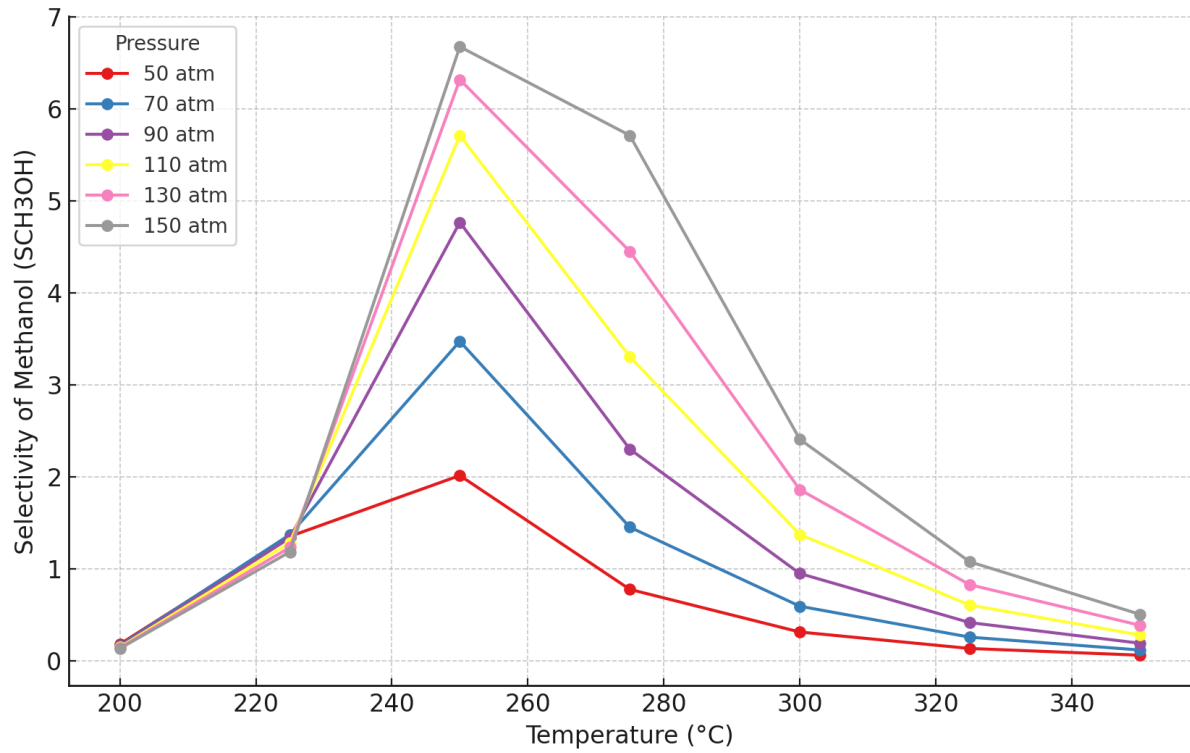


Figure 23: Selectivity of methanol vs temperature at various pressure for comparison with reference [81]

The simulation results obtained using the Vanden Bussche–Froment kinetic model for our process condition are also compared with the reference Graaf kinetic model [75]. **Table. 12** briefly compares the reactor outlet stream using the VBF and Graaf kinetic model.

Table 12: Comparison of reactor outlet stream using different kinetic models

Parameters	Vanden Bussche–Froment	Graaf kinetic [75]
Mole Flows (kmol/hr)	878.99	1019
CO (kmol/hr)	15.08	16
H ₂ (kmol/hr)	584.24	849
CH ₄ O (kmol/hr)	43.54	44
CO ₂ (kmol/hr)	194.64	69
H ₂ O (kmol/hr)	41.48	42
Single pass Conversion %	16.4	32.3
Overall Conversion %	97.4	97.6

The chosen Vanden-Bussche-Froment kinetic model is able to capture the influence of inlet temperature, pressure, and feed composition in an effective manner (Fig. 24-25). The

differences in single-pass conversion for the VBF and Graaf kinetics models in methanol synthesis from CO_2 hydrogenation highlights the importance of reaction kinetics and the reactor design, especially when considering recycling unreacted gases. This difference in the conversion could be due to equilibrium limitations, different reaction rates, and pathways considered in the kinetic model. Despite the differences in single-pass conversion, the overall conversions for both kinetics are quite high and very close to each other. This high overall conversion is achieved through the recycling of unreacted gases back into the reactor system. A sensitivity analysis of reactor process variables is performed to analyze their effects on the methanol production rate. From Fig. 24-25 it is concluded that with the increase in the reactor temperature, the methanol production rate is increased but up to a specific limit after that, a further increase in temperature decreases the production rate. In case of an increase in reactor pressure, the production rate increases nonlinearly, confirming the exothermic nature of the reaction.

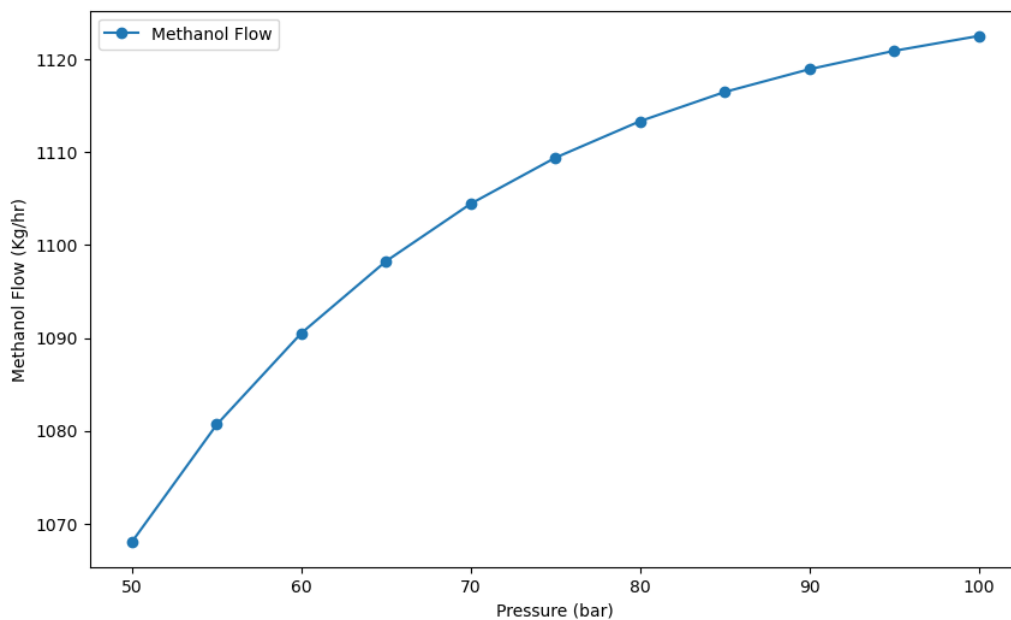


Figure 24: Effect of reactor operating pressure on methanol production rate at constant 250°C

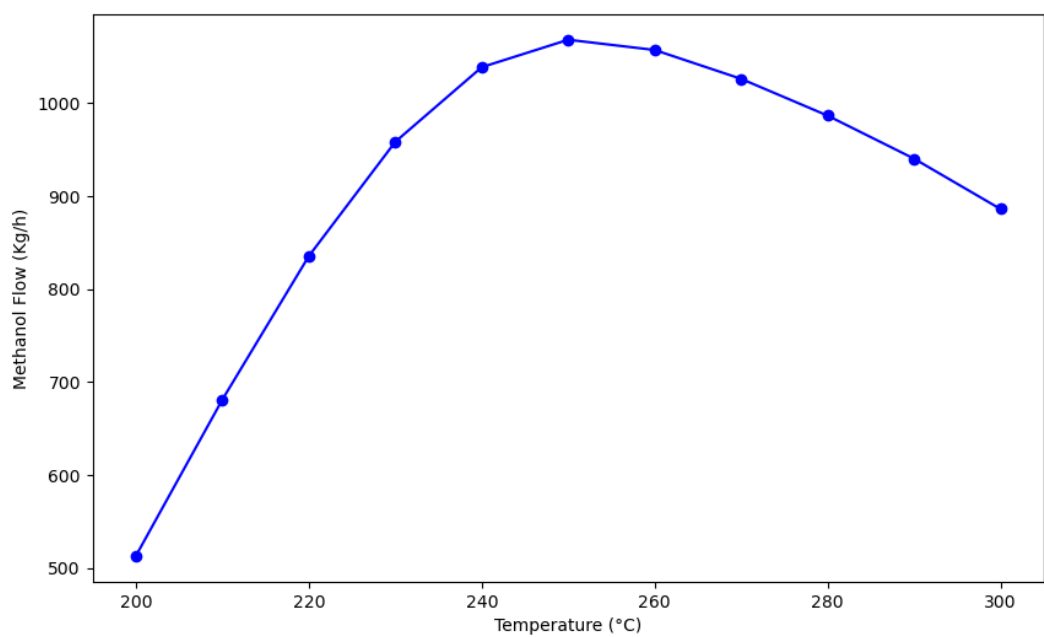


Figure 25: Effect of reactor operating temperature on methanol production rate at constant 50 bar pressure

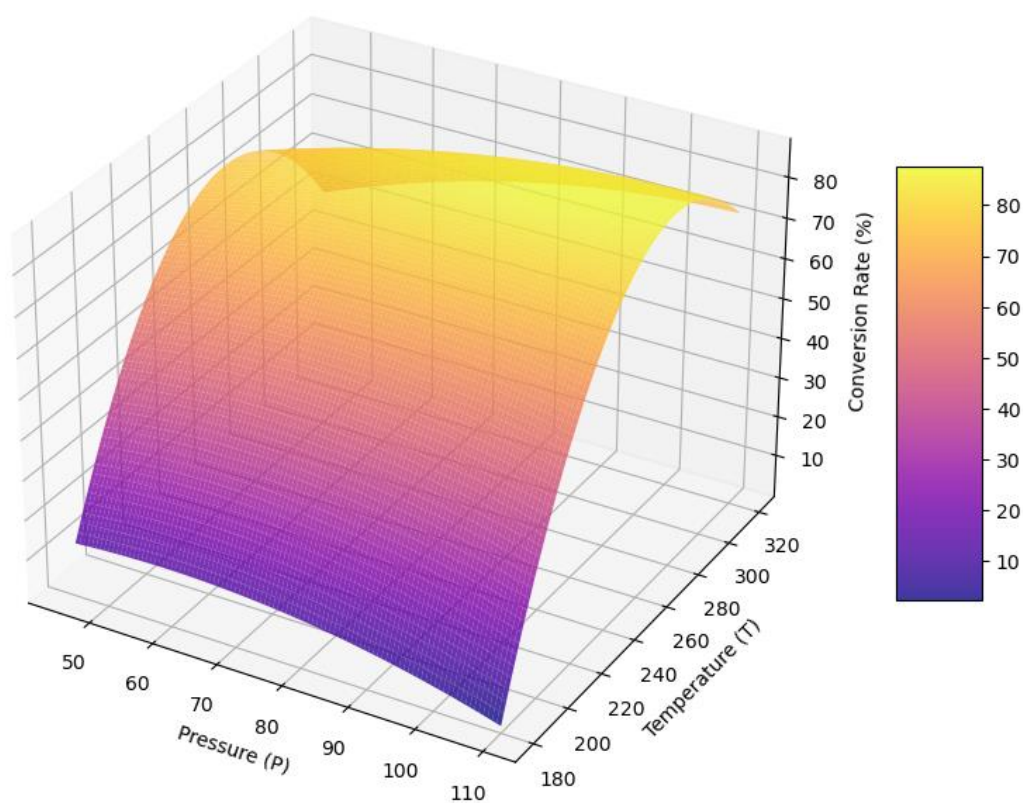


Figure 26: Surface plot of temperature-pressure and conversion

4.2 Feature Importance

To visualize the effect of the selected input variables on the methanol production rate, purity, and energy requirement, a Pearson correlation matrix of the input and output pairs is formulated, as shown in **Fig.27**. It is a statistical method that quantifies how strongly variables are related to each other. The correlation values range from -1 to 1, where 1 indicates a perfect positive correlation (Dark green boxes), -1 indicates a perfect negative correlation (Dark purple boxes), and 0 indicates no or minimum correlation between input-output variables (light pink boxes).

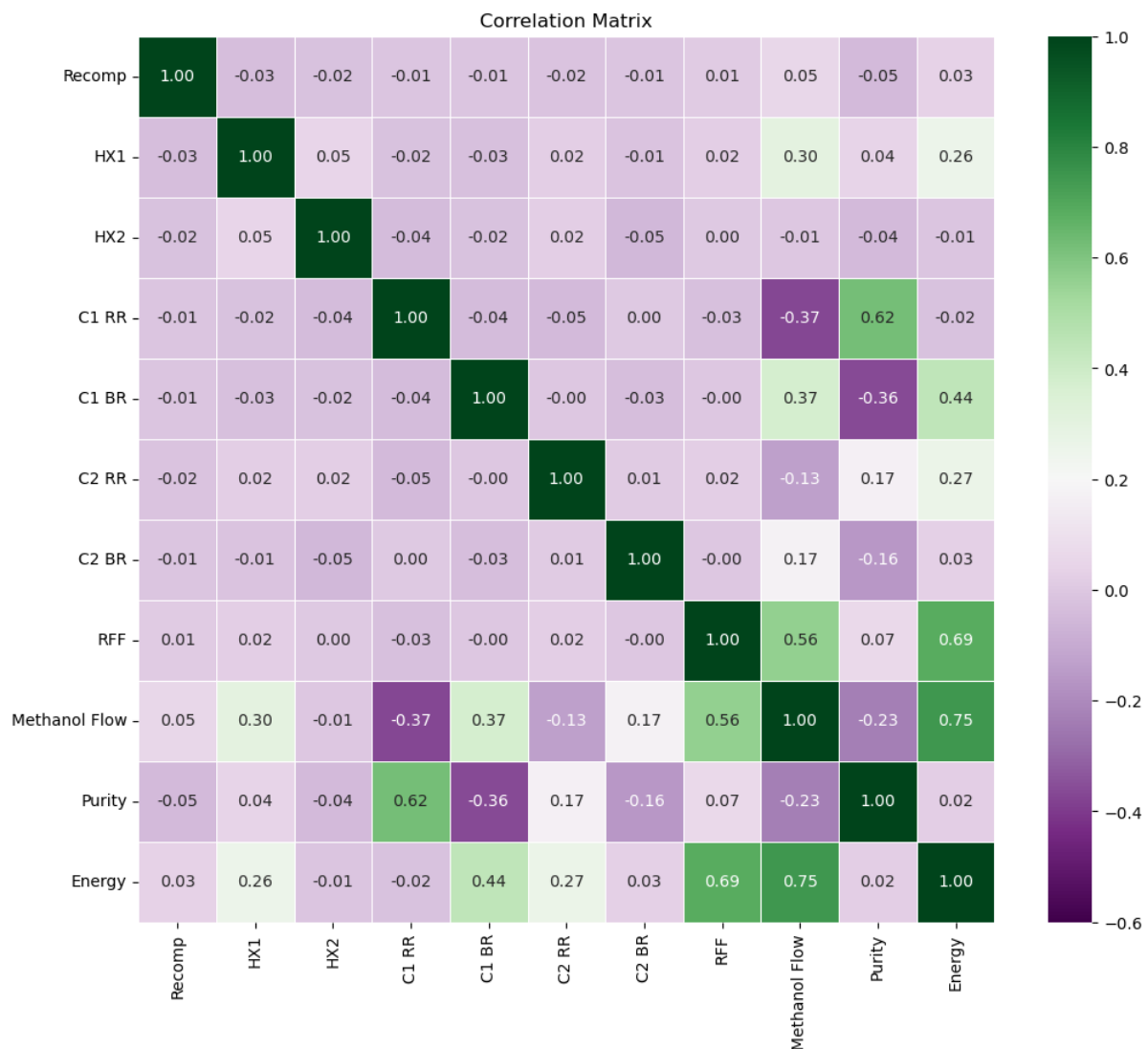


Figure 27: Pearson correlation matrix for input-output variables

From **Fig.27**, it can be concluded that the selected process variables strongly impact the prediction of target variables of interest. For predicting the methanol production rate, reactor operating temperature, pressure, recycle flow fraction (RFF), and the distillation column operating variables have a strong dependency. For final product purity, distillation column operating conditions have a strong dependency, and for energy prediction, recycle flow fraction (RFF), reactor temperature, distillation column-1 boil-up ratio, and column-2 reflux ratio are the most significant factors. These relationships indicate that carefully controlling these variables is crucial for efficient operation.

4.3 Analyze the effect of sampling size and techniques.

For the analysis of the effect of sample size and sampling technique on the performance of the surrogate model, the Support vector regressor (SVR) ML model is trained on the dataset mentioned in **Table.9**. The model's performance is evaluated on the coefficient of determination (R^2) and mean-squared error (MSE) performance metrics. The results of comparing different sampling techniques and the sample size on the model's performance are shown in **Fig.28-29**. From the figures, it can be concluded that the model performance improves for all three selected sampling techniques as the sample size increases from 100 to 2000. This is demonstrated by the decreasing trend in MSE and the increase in R^2 with the increase in sample size. Also, large sample sizes lead to more accurate models due to a greater amount of information available to capture the patterns in the data. When comparing different sampling techniques, the LHS performs significantly better for all the sample sizes, followed by SOBOL sampling. This suggested the LHS is more effective when dealing with smaller datasets. However, as the sample size increases, the choice of sampling technique becomes less critical in terms of the model's performance.

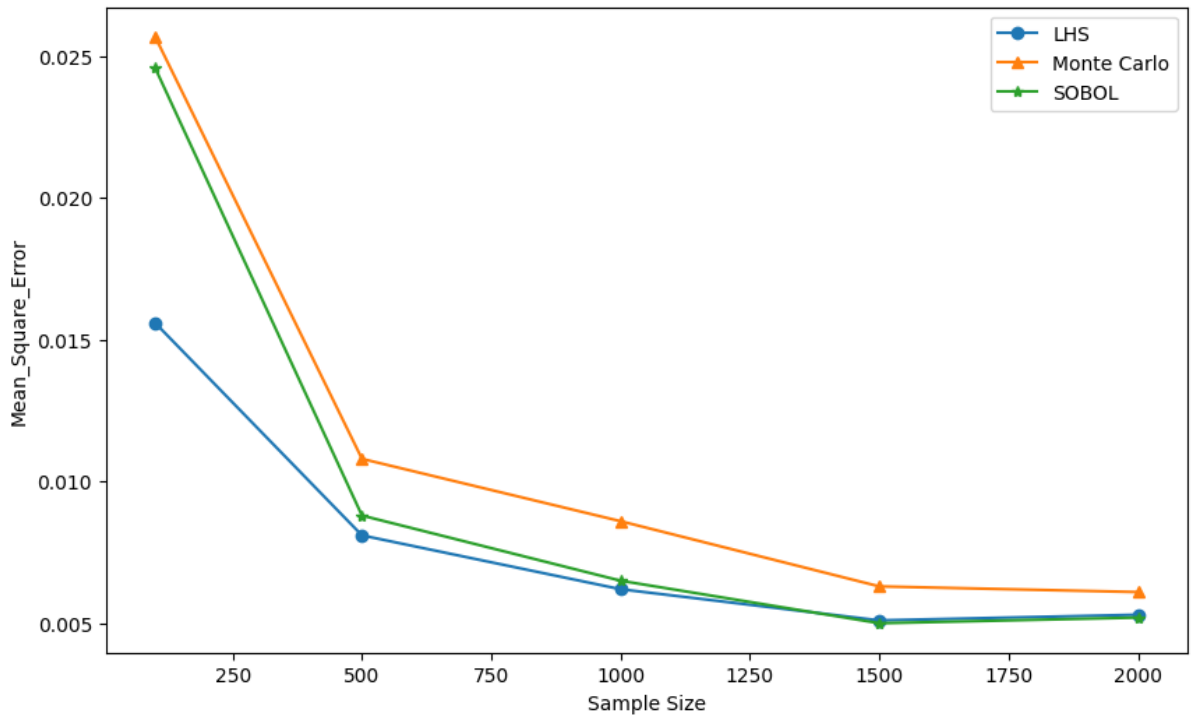


Figure 28: Effect of different space filling techniques and sample size on SVR model MSE in predicting production rate

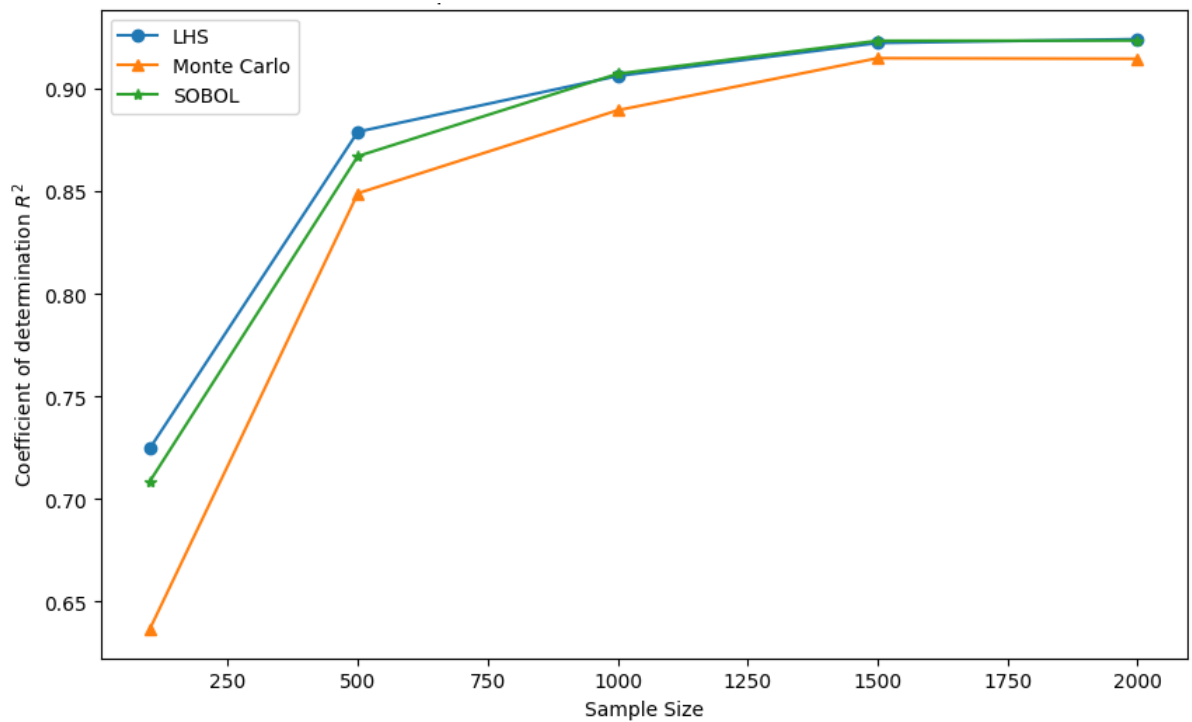


Figure 29: Effect of different space filling techniques and sample size on SVR model R^2 in predicting production rate

4.4 Model Training and screening

To prepare the surrogate model that can accurately predict methanol production rate, energy requirement, and final product purity. Support vector regressors (SVR), random forests, decision trees, and XG-boost regressors are trained on the 2000 data set prepared with the LHS technique. About 20 percent of the data is used for model validation, and 80 percent of the data is used for training. First, the trained ML models undergo validation using the validation set, and after that, their performance is evaluated on a separate test data set. A deep learning model, like an artificial neural network with a ReLU activation function with two hidden layers and eight neurons in each layer, is also trained to compare its performance with the ML models. Fig.30 shows the actual simulation output for the selected target variables and Fig.31-32 shows the fitting of trained ML and DL models on the test data set, and Fig.33 provides the comparison of actual simulation output and the predicted output of ANN and Xg-Boost models. From the model results Table.13, it is concluded that the gradient boosting model XG-Boost and the deep learning model Artificial Neural Network (ANN) exhibited superior performance over simple ML models like SVR, Random Forest, and Decision Tree in predicting energy requirement, methanol flux, and purity. The XG-Boost showed excellent accuracy and low error rates across all metrics (R^2 , MAE, MSE), outperforming simple models in both validation and test sets. However, the ANN emerged as the most effective model, achieving the highest R^2 scores and lowest error metrics, particularly in the test set's energy requirement and methanol flux predictions. This analysis underscores the robustness of gradient boosting and the exceptional capability of deep learning in handling complex and non-linear data, with simple models like Random Forest and SVR showing commendable but comparatively less impressive results. The results of model performance on both validation and test data sets are summarized in Table. 13.

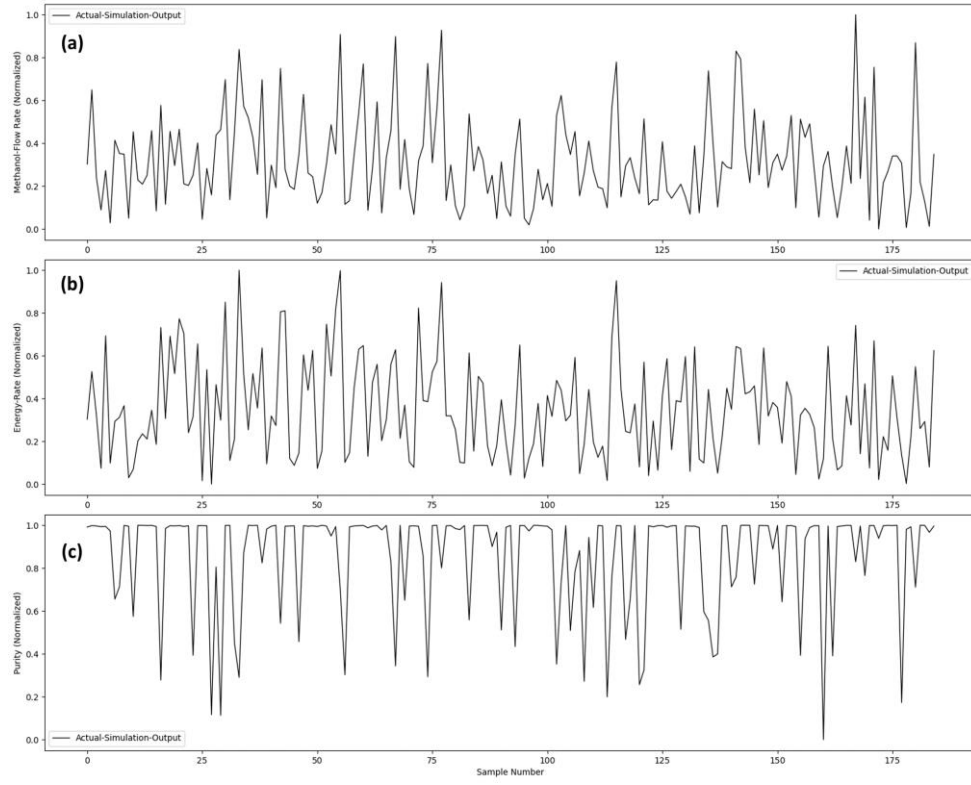


Figure 30: Actual simulation output for selected target variables (a) Production rate (b) Energy Consumption rate, and (c) purity of final product

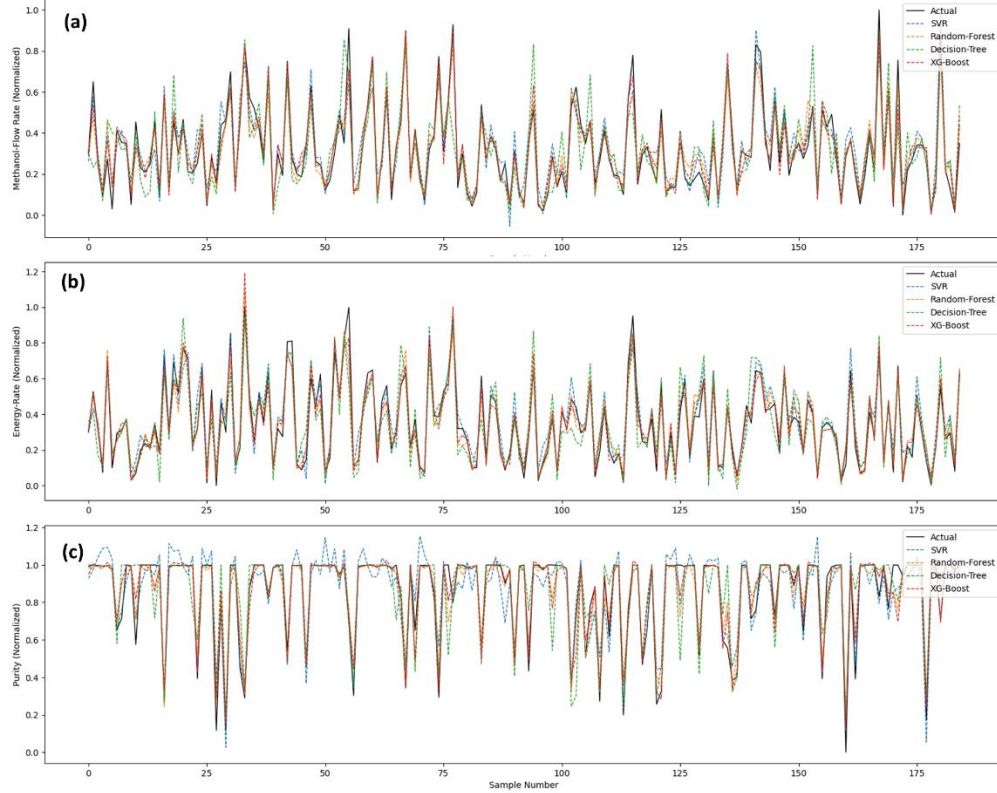


Figure 31: ML models (SVR, Random-Forest, Decision-Tree, & XG-Boost) results on test data sets (a) Production rate (b) Energy Consumption rate, and (c) purity of final product

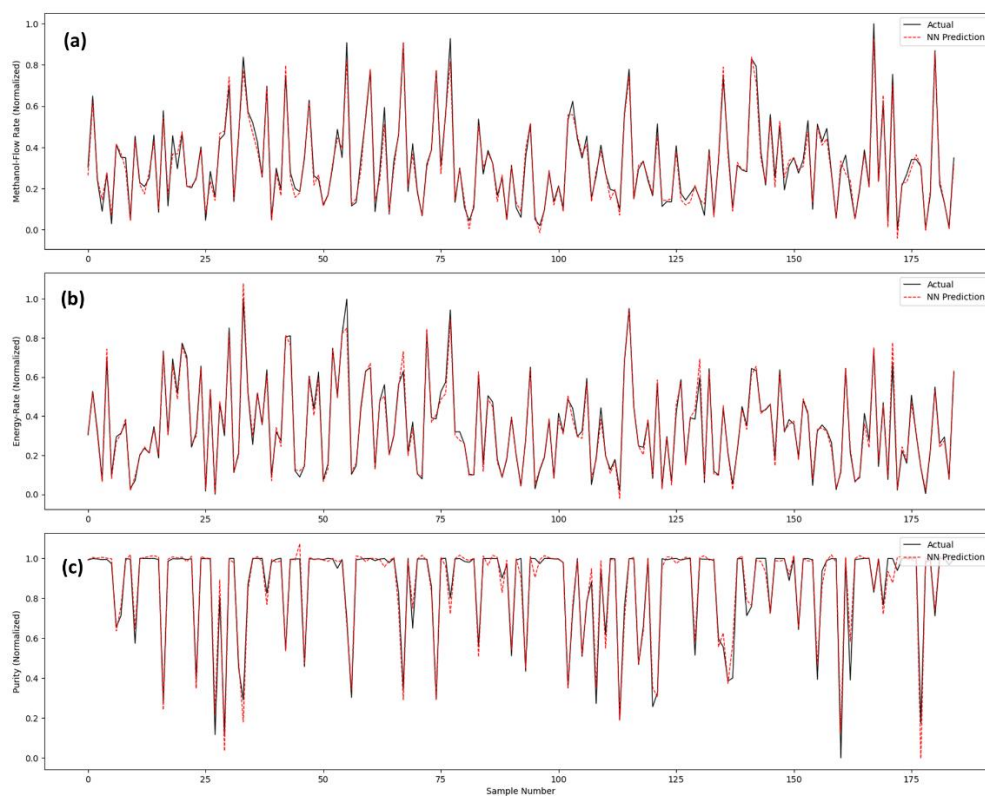


Figure 32: DL-model (ANN) results on test data set (a) Production rate (b) Energy Consumption rate, and (c) purity of final product

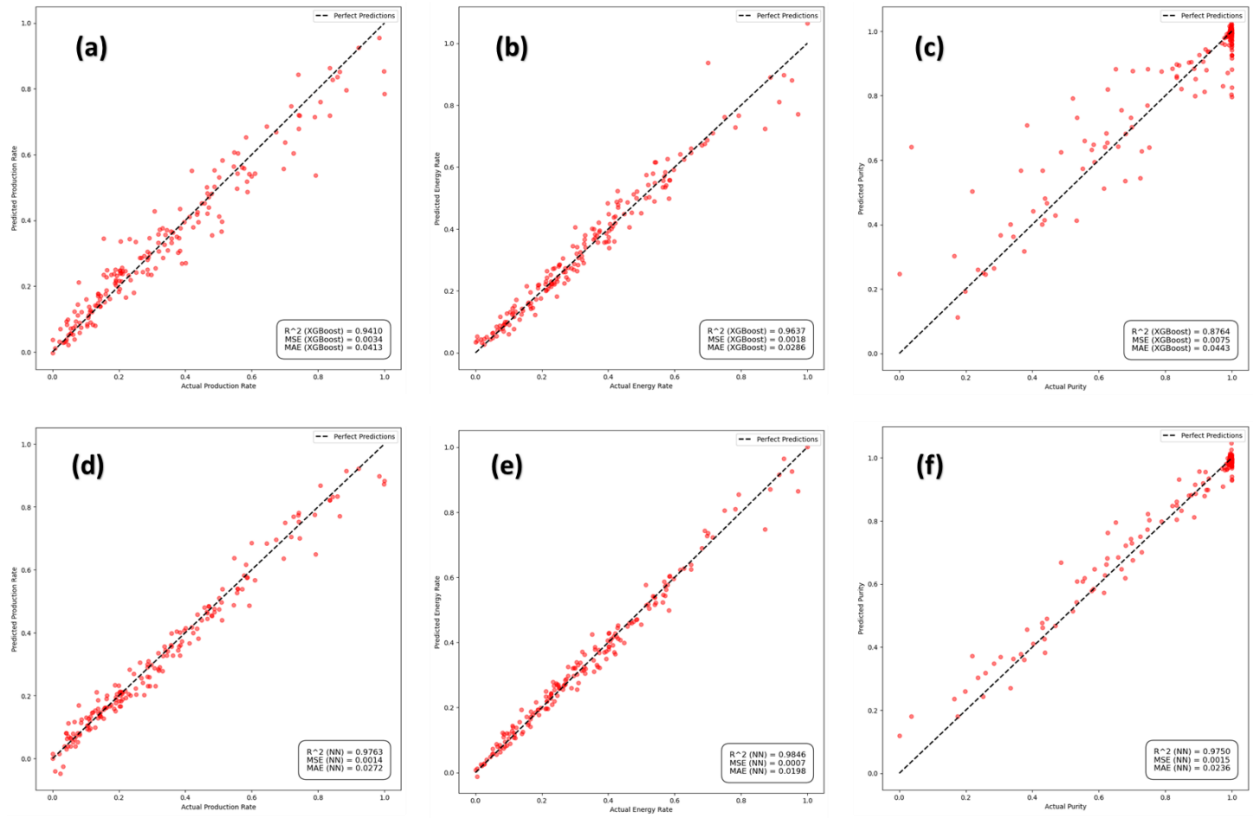


Figure 33: Comparison of actual vs predicted results of ANN and XG-Boost models. (a) Xg-Boost model production rate prediction, (b) Xg-Boost model energy rate prediction, (c) Xg-Boost model purity prediction. (d) ANN model production rate prediction, (e) ANN model energy rate prediction, (f) ANN model purity prediction.

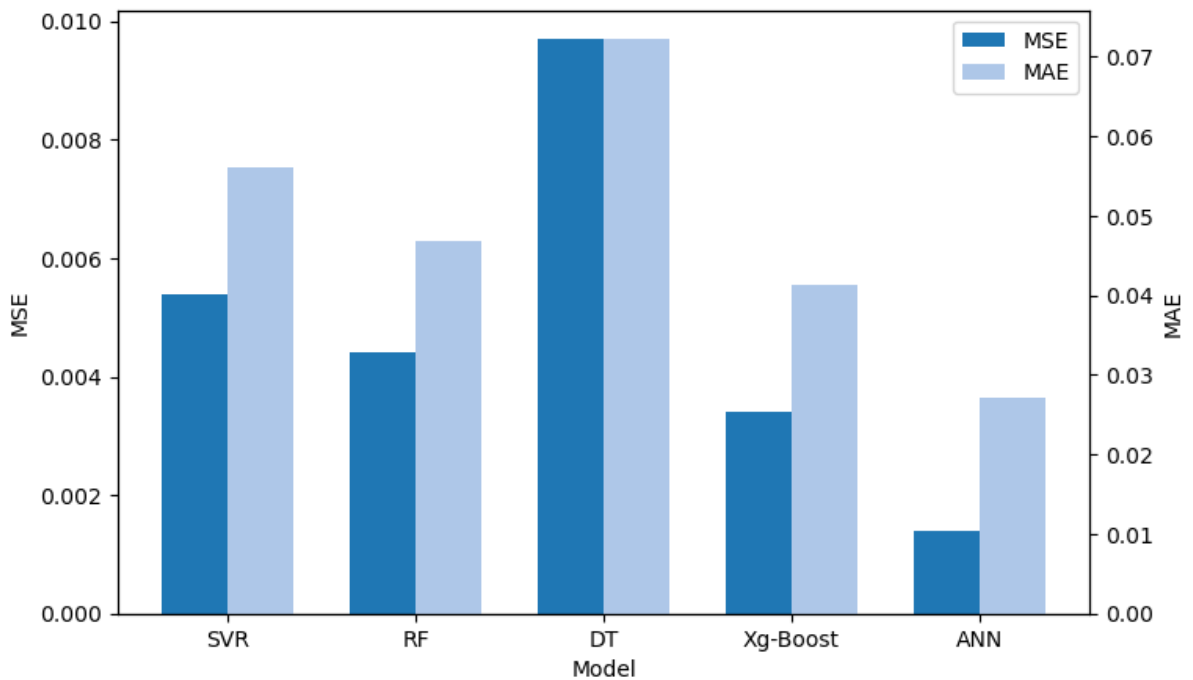


Figure 34: ML and DL model performance comparison for predicting production rate on test dataset

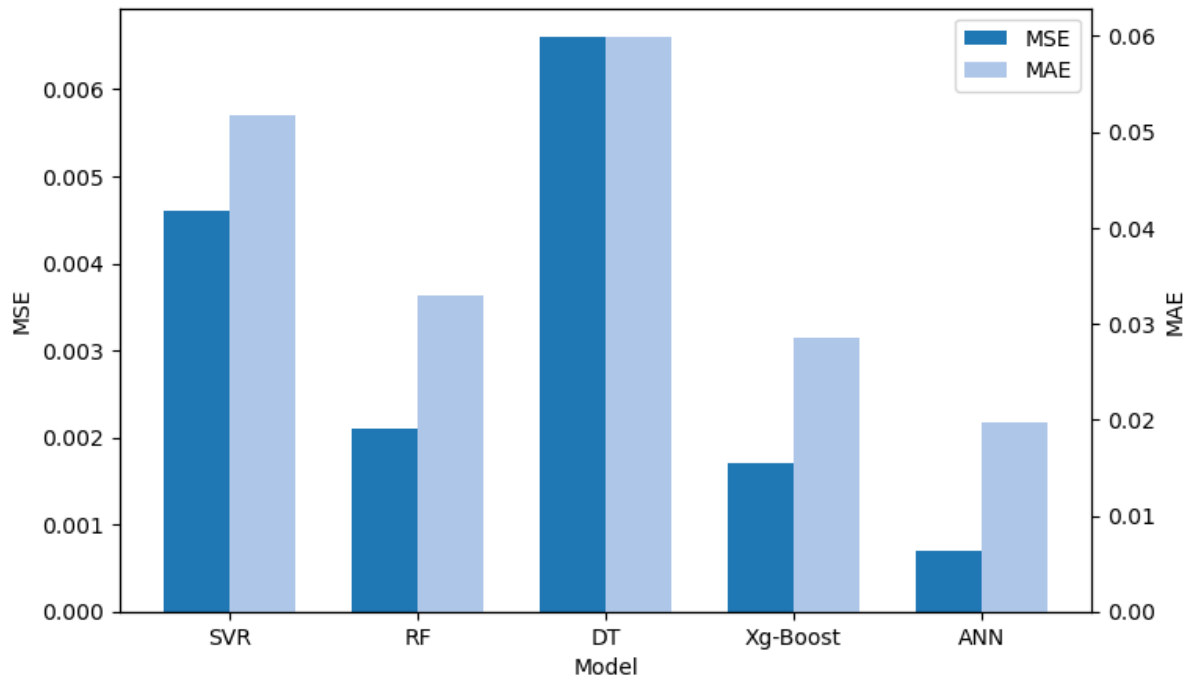


Figure 35: ML and DL model performance comparison for predicting energy consumption rate on test dataset

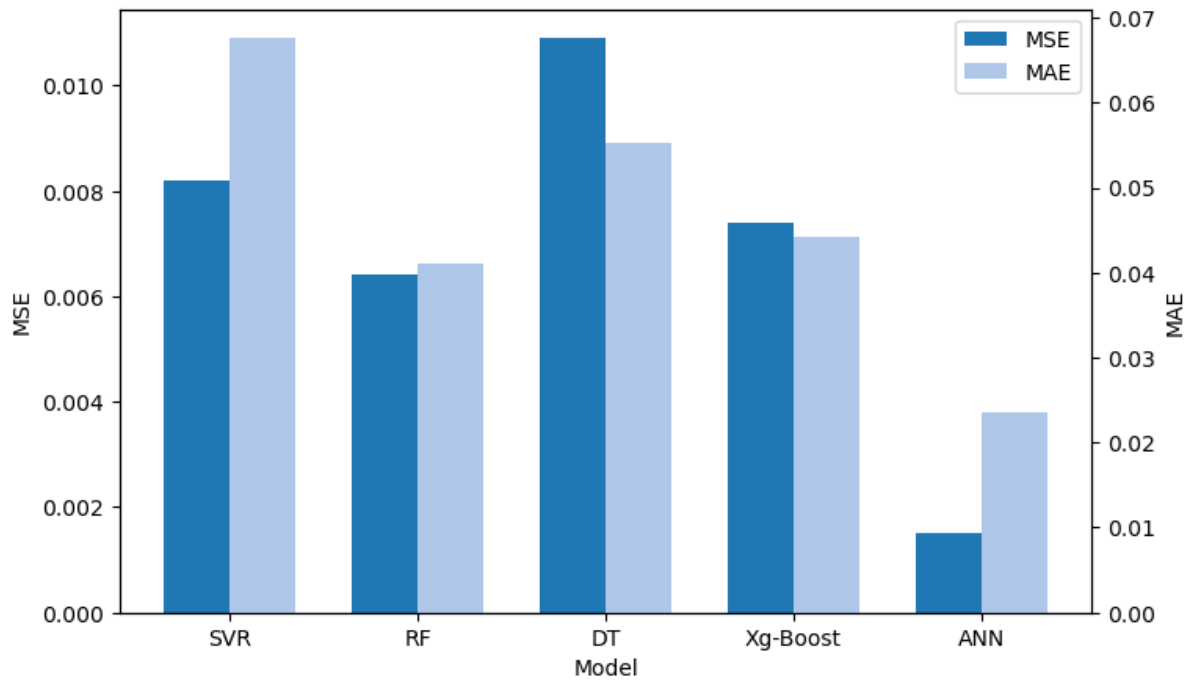


Figure 36: ML and DL model performance comparison for predicting final product purity on test dataset

Table 13: Summary of ML and DL model performance for production rate, energy requirements, and purity predictions

Models		Energy Requirement			Methanol Flow			Purity		
		R ²	MAE	MSE	R ²	MAE	MSE	R ²	MAE	MSE
Training Set										
ML models	SVR	0.9385	0.0472	0.0032	0.9339	0.0461	0.0031	0.9362	0.0529	0.0040
	Random Forest	0.9935	0.0135	0.0003	0.9889	0.0172	0.0006	0.9881	0.0145	0.0007
	Decision Tree	1.000	0.0000	0.0000	1.000	0.000	0.000	1.000	0.0000	0.000
	XG-boost	0.9999	0.0014	0.0001	0.9999	0.0021	0.0001	0.9999	0.0014	0.0001
DL models	ANN	0.9871	0.0188	0.0006	0.9750	0.0250	0.0011	0.9824	0.0192	0.0009
Validation Set										
ML models	SVR	0.9098	0.0517	0.0043	0.9060	0.0519	0.0043	0.9094	0.0623	0.0064
	Random Forest	0.9516	0.0357	0.0023	0.9198	0.0436	0.0036	0.9029	0.0438	0.0069
	Decision Tree	0.8519	0.0596	0.0071	0.8174	0.0661	0.0083	0.8455	0.0529	0.0109
	XG-boost	0.9689	0.0279	0.0015	0.9439	0.0362	0.0025	0.9219	0.0414	0.0055
DL models	ANN	0.9821	0.0215	0.0008	0.9749	0.0249	0.0011	0.9762	0.0232	0.0016
Test Set										
ML models	SVR	0.9028	0.0517	0.0046	0.9070	0.0560	0.0054	0.8641	0.0676	0.0082
	Random Forest	0.9554	0.0329	0.0021	0.9229	0.0469	0.0044	0.8941	0.0411	0.0064
	Decision Tree	0.8628	0.0599	0.0066	0.8327	0.0722	0.0097	0.8197	0.0553	0.0109
	XG-boost	0.9636	0.0285	0.0017	0.9409	0.0412	0.0034	0.8763	0.0442	0.0074
DL models	ANN	0.9846	0.0198	0.0007	0.9763	0.0272	0.0014	0.9750	0.0236	0.0015

4.5 Process Optimization

In this section, the optimization of the process variable of the methanol production process is discussed. From section 4.4, it is concluded that the Deep-learning-based ANN model emerged as the most accurate model for describing the energy consumption rate, product purity, and production rate. So, it has been incorporated into the optimization problem formulation to minimize the energy consumption rate and maximize the product purity and production rate. The developed ANN model is trained on eight different non-linear variables, resulting in a non-linear programming (NLP) optimization problem. The objective function and the constraints of the optimization problem are as follows:

$$\min f(x) = E_T(x) - M_{CH_3OH}(x) - x_{CH_3OH}(x) \text{ --- Eq. (4.5.1)}$$

$$s. t \quad LB_i \leq x_i \leq UB_i \quad i = 1, 2, 3, \dots, N$$

Where; $f(x)$, is the objective function calculated based on the vector of input variables x mentioned in Table.8. $E_T(x)$, $M_{CH_3OH}(x)$ and $x_{CH_3OH}(x)$ represents the ANN-trained model for the final product purity, methanol production, and energy consumption rate, respectively. LB_i and UB_i represents the upper and lower bounds of the input variables, ensuring the input variables are manipulated within the specified range. The upper and lower bounds of the input variables are summarized in **Table.8**. The optimization problem was solved in the Python environment with two optimization algorithms: Sequential least square quadratic programming (SLSQP) and Differential Evolution algorithm (DEA). **Table.13** summarizes the optimization results for the methanol production process using the DEA and SLSQP algorithms.

Table 14: Summary of Methanol production optimization results

Variables	Units	Nominal Values	Base Case	Optimization (Eq. 4.5.1)	
				DEA	SLSQP
T_1	°C	75	75	99	70.7
P_1	bar	250	230	251.3	240.4
T_2	°C	90	87	86.9	86.176
RR_{C1}	—	7	1.11	2.104	11.1094
BR_{C1}	—	7.9	0.8	0.84	14.432
RR_{C2}	—	7	1.11	1.063	7.3
BR_{C2}	—	7.9	0.6	7.756	11.4
RFF	—	0.75	0.90	0.936	0.712
M_{CH_3OH}	kg/hr	653.787	759.175	1010.14	589.102
x_{CH_3OH}	—	0.99	0.97	0.99	0.99
E_T	kW	13600.8	5606.95	6604.94	12429.3
SER	kWh/kg	16.9	7.38	6.53	21.09
Objective Function		-1	-1.320	-1.517	-0.908
CPU Time		-	-	18.23 min	0.009 min
ML-Models Predictions					
M_{CH_3OH}	kg/hr	-	-	963.98	575.339
x_{CH_3OH}	—	-	-	0.989	0.96
E_T	kW	-	-	6930.23	13972.34

The fifth and sixth columns represent the results obtained after applying DEA and SLSQP algorithms on Eq. 4.5.1. The results reveal that the optimized variables using DEA algorithm provide acceptable optimization results in terms of maximum production rate and competitive purity with lower energy requirements as compared to the SLSQP. The energy consumption rate is approximately 47% lower, while the production rate experiences a boost of around 71% compared to the SLSQP-optimized results. These findings also suggest that the final solution using SLSQP did not reach the global optimum but converged to a local optimum point and required multiple initial guesses to reach the global optimum solution. In comparison with base case results, the production rate is more than 33% but at the expense of 18% higher energy requirements. This finding reveals a trade-off between the production rate, purity, and energy consumption, where achieving a higher production rate and purity requires an increase in energy

input. The optimization results derived from the developed surrogate models are also compared with actual simulation outputs to assess their effectiveness. Although surrogate models substantially reduce computational costs compared to first-principle models, a compromise in accuracy is evident, especially in the energy consumption predictions, as shown in Table 14. This comparison underscores the need for enhancing model precision, and to address this, refining the hyperparameters or incorporating more data points during model training is recommended to reduce discrepancies between the model outputs and actual simulation results.

Chapter 5. Conclusion and Future Work

5.1 Conclusion

This study presents the ML-based surrogate modeling of methanol production using data generated from the Aspen simulation software. The main focus was to prepare models that can predict the plant energy requirements, production rate, and final product purity under variable operating conditions and to optimize these variables with efficient exploration of the wide range of operating conditions and design spaces. This research also compares the effect of sampling techniques (LHS, SOBOL, Monte-Carlo) and sampling size on the performance of the surrogate model. The findings reveal that LHS consistently outperforms Sobol and Monte Carlo in smaller sample sizes due to its superior uniform distribution of input variables. However, as the sample size increases, the performance gap narrows, indicating less dependency on the sampling technique. Notably, while Sobol's performance initially ranks between Monte Carlo and LHS in smaller datasets, it approaches the effectiveness of LHS in larger sample sizes. This trend suggests that while LHS is preferable for smaller datasets, the choice of sampling method becomes less important in larger datasets.

In order to prepare accurate models of energy requirement, production rate, and product purity, a range of ML algorithms, including Support Vector Regression (SVR), Random Forest, Decision Tree, and XG-Boost, were employed. Their performance was compared against a deep learning-based Artificial Neural Network (ANN) model. This comparative analysis was conducted on a dataset comprising 2,000 data points generated via the Latin Hypercube Sampling (LHS) technique. The results reveal that the gradient boosting XG-Boost and the deep learning ANN models significantly outperform simpler models like SVR, Random Forest, and Decision Tree in predicting energy requirements, methanol flux, and purity. XG-Boost exhibited outstanding accuracy and low error rates across all evaluation metrics, but at the same time, the

ANN model demonstrated superior performance, particularly in predicting energy requirement and production rate, evidenced by the highest R^2 scores and lowest error metrics in the test set. The ANN-based trained models achieved $R^2 > 0.95$ across all the target variables and were well generalized. Therefore, it has been incorporated into the optimization problem formulation to minimize the energy consumption rate and maximize production and purity. The non-linear optimization algorithms, SLSQP and DEA, are employed for the process variable optimization. The SLSQP algorithm fails to optimize the process variable efficiently due to its inherent nature to converge to the local optimum solution and require multiple initial guesses. The DEA, which is a global optimization solver, gives acceptable optimization results. The optimized variable with DEA algorithms gives a significantly 33% higher production rate and competitive purity of the final product but at the expense of 18% higher energy requirement from the base case conditions; this reveals the trade-off between production rate, purity, and energy requirements.

The comparative analysis presented in this research highlights the advanced capabilities of gradient boosting and deep learning models in handling complex, non-linear datasets, with simpler models showing respectable yet less accurate results. The insights from this study are also crucial for the selection of appropriate modeling and sampling techniques for enhanced precision and effectiveness of the surrogate models.

5.2 Recommendations and Future Work

The data-driven optimization strategy discussed in this work is generalized and can be used for the optimization of any industrial process. The developed surrogate model can be trained on industrial process data to develop a hybrid model that can facilitate a more accurate prediction of the real-world output of the process plant. We can also incorporate variations of ambient conditions around the plant during the training phase; in that way, we can analyze the effect of ambient conditions on the plant production rate and have a more realistic output from the model. Future work intends to develop surrogate models for different chemical product

processes that can be produced from the PtX concept and to study the novel integrated process optimization. The surrogate model can facilitate the provision of accurate output from the different processes under uncertain conditions of raw material availability, i.e., carbon dioxide and green hydrogen. In that way, we can maximize the total profit under uncertain market conditions while satisfying the overall facility's demand, quality, safety, and environmental constraints.

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Appendix-1: Sampling Codes

Latin Hyper Cube

```
# Step-1: Import necessary libraries and define the ranges
import numpy as np
from pyDOE2 import lhs

# Step-2: Define the ranges for the variables
ranges = np.array([
    [50, 100], # Repump
    [200, 300], # HX1
    [85, 95],   # HX2
    [1, 15],    # C1 RR
    [0.8, 15],  # C1 BR
    [1, 15],    # C2 RR
    [0.6, 15],  # C2 BR
    [0.5, 0.99], # RFF
])

# Step 3: Generate the samples using LHS
n_samples = 2000
n_variables = ranges.shape[0]
samples = lhs(n_variables, samples=n_samples)

# Step 4: Scale the samples to the given ranges
scaled_samples = np.zeros(samples.shape)
for i in range(n_variables):
    scaled_samples[:, i] = samples[:, i] * (ranges[i, 1] - ranges[i, 0]) + ranges[i, 0]

# Step 5: Create a DataFrame
import pandas as pd

column_names = ['Repump', 'HX1', 'HX2', 'C1 RR', 'C1 BR', 'C2 RR', 'C2 BR', 'RFF']
df = pd.DataFrame(scaled_samples, columns=column_names)

# Step 6: Save the DataFrame to a excel file
excel_filename = "New_LHS_samples_2000.xlsx"
df.to_excel(excel_filename, index=False, engine='xlsxwriter')

print(f"Data points saved to {excel_filename}")
```

SOBOL Sequence:

```
# Step 1: Import the necessary libraries
import sobol_seq
import numpy as np

# Step 2: Define the ranges for the variables
n_samples = 2000
n_variables = 10
samples = sobol_seq.i4_sobol_generate(n_variables, n_samples)

ranges = np.array([
    [50, 100], # Repump
    [200, 300], # HX1
    [85, 95],   # HX2
    [1, 15],    # C1 RR
    [0.8, 15],  # C1 BR
    [1, 15],    # C2 RR
    [0.6, 15],  # C2 BR
    [0.5, 0.99], # RFF
])

# Step 3: Scale the samples to the given ranges
scaled_samples = np.zeros(samples.shape)
for i in range(n_variables):
    scaled_samples[:, i] = samples[:, i] * (ranges[i, 1] - ranges[i, 0]) + ranges[i, 0]

# Step 4: Convert the samples to a pandas DataFrame
import pandas as pd

column_names = ['Repump', 'HX1', 'HX2', 'C1 RR', 'C1 BR', 'C2 RR', 'C2 BR', 'RFF']
df = pd.DataFrame(scaled_samples, columns=column_names)

# Step 5: Write the DataFrame to an Excel file
excel_filename = "sobol_samples_2000.xlsx"
df.to_excel(excel_filename, index=False)

print(f"Data points saved to {excel_filename}")
```

Mote-Carlo Sampling

```
# Step 1: Import the necessary libraries
import numpy as np
import pandas as pd

# Step 2: Define the ranges for your variables
ranges = np.array([
    [50, 100], # Repump
    [200, 300], # HX1
    [85, 95],   # HX2
    [1, 15],    # C1 RR
    [0.8, 15],  # C1 BR
    [1, 15],    # C2 RR
    [0.6, 15],  # C2 BR
    [0.5, 0.99], # RFF
])

# Step 3: Generate random samples uniformly for each range

n_samples = 100
n_variables = ranges.shape[0]

samples = np.random.rand(n_samples, n_variables)
scaled_samples = np.zeros(samples.shape)
for i in range(n_variables):
    scaled_samples[:, i] = samples[:, i] * (ranges[i, 1] - ranges[i, 0]) + ranges[i, 0]

# Step 4: Convert the samples to a pandas DataFrame
column_names = ['Repump', 'HX1', 'HX2', 'C1 RR', 'C1 BR', 'C2 RR', 'C2 BR', 'RFF']
df = pd.DataFrame(scaled_samples, columns=column_names)

# Step 5: Write the DataFrame to an Excel file
excel_filename = "monte_carlo_samples_100.xlsx"
df.to_excel(excel_filename, index=False)

print(f>Data points saved to {excel_filename}")
```